The Making of Dominant Vehicle Currencies: Evidence from DeFi

August 21, 2026

Abstract

We use 472 million swaps to observe vehicle currencies in transaction-level routes on decentralised exchanges. Between matched January–June periods in 2024 and 2026, stablecoins' route-count share among native and stable vehicles rises from 16.9% to 42.3%. Opposing substitutions offset within continuing source–destination pairs; reallocation and pair entry and exit generate the increase. The vehicle used at pair entry predicts later allocation. After persistent stablecoin-bridge support, first-month adoption is 42.6% with shallow depth and 84.1% once depth reaches one-tenth of WETH depth. Exact price competition changes venues much more often than vehicles. New trading relationships and two-leg liquidity jointly shape vehicle dominance.

1 Introduction

Every cross-border payment between two currencies needs an institution willing to receive one and deliver the other. When a payment provider lacks access to the destination payment system and inventory of the destination currency, the BIS Project Nexus design assigns that task to a third-party foreign-exchange provider (Bank for International Settlements Innovation Hub, 2024). Project Rialto goes one step further: its experimental cross-border system uses a third currency when direct exchange is unavailable (Bank for International Settlements Innovation Hub, 2025). The practical question is which currency connects a thin trading relationship and what liquidity makes that connection viable.

The same problem underlies the dominance of the U.S. dollar. The dollar appears on more than 85% of foreign-exchange transactions, far more often than the United States' share of cross-border economic activity would suggest (Somogyi, 2026). Its use as a vehicle between other currencies concentrates trading in dollar markets; deeper markets lower trading costs and reinforce the dollar's intermediary role (Krugman, 1980; Kiyotaki and Wright, 1989; Flandreau and Jobst, 2009). This feedback explains persistence after a vehicle has become dominant. It says less about how a challenger gains share when new trading relationships and new market technologies appear.

Vehicle dominance depends on which intermediary connects a source to a destination and how much is traded across different relationships. Aggregate dominance can therefore change even when the vehicle used inside every continuing relationship remains unchanged. Measuring these two channels requires transactions that attach each intermediary to the exchange between the final source and destination.

Standard foreign-exchange records report turnover in each currency pair without linking the legs of the customer's exchange.[1] A yen–euro exchange routed through dollars appears as separate yen–dollar and dollar–euro trades. Somogyi (2026) infers vehicle activity from currency-pair volume responses around non-overlapping holidays. In decentralised exchange, by contrast, the pool calls inside one blockchain transaction preserve their execution order. We use these records to observe which asset connects the source to the destination.

We call the ordered source and destination the *endpoint pair*, shortened hereafter to *pair*. Each directed pool execution is a *leg*, and a *route* is the observed ordered sequence of legs. A route may contain one leg, several sequential legs, or a split across pools. We reconstruct connected routes from 472 million swaps on eight Ethereum automated market maker (AMM) deployments between February 2020 and June 2026. The resulting token sequences identify the pair, every intermediary, and whether the route spans exchanges. They reveal vehicle use directly, avoiding inference from turnover in separate legs.

Stablecoins, including USDC, USDT, DAI, and other pegged tokens, become substantially more prominent vehicles during the sample. We compare the same January-through-June dates in 2024 and 2026 because the 2026 data end on June 30. Stablecoin dominance rises from 16.9% to 42.3% by route count and from 32.7% to 76.5% by dollar-weighted volume. Within pairs active in both periods, positive and negative stable-for-native substitutions offset in aggregate. Trading shifts toward continuing pairs that already rely more on stablecoins, and pairs present in only one comparison period supply the remaining increase.

The rotation is not monotonic. Native assets regain share in 2023 and 2024 before stablecoins recover the dollar-weighted lead in 2025. This reversal makes a smooth technology trend an incomplete explanation for the 2024–2026 change.

The first routes of a new pair strongly predict its vehicle use over the next 120 days, including after removing pairs with WETH at an endpoint and controlling for entry size and route structure. Persistent bridge support also commonly precedes first use of the supported stablecoin. First-month adoption is 42.6% when stablecoin depth is below one-tenth of WETH depth and 84.1% once it reaches that threshold. The gap survives controls for prior WETH activity and remains among pairs with two non-stablecoin endpoints. Among eventual adopters, deposited capital on the weaker leg rises before the first observed stablecoin route. These descriptive associations place vehicle formation where new trading relationships meet two-leg liquidity; they do not identify an exogenous liquidity-supply effect.

Directly observed routes change the interpretation of vehicle-currency dominance. A large aggregate rotation can arise because the trading network expands around a challenger, even when continuing pairs show little net substitution. Exact pre-transaction prices often favour another

[1] For example, see the BIS 2025 Triennial Survey reporting guidelines and templates.

venue, while opening other vehicles and the direct path lowers stablecoin vehicle share by only 1.2 pp. This formation channel complements the holiday-based inference of Somogyi (2026) and the historical network evidence of Flandreau and Jobst (2009). The persistence, price, and depth results connect that channel to classical theories in which vehicle use and liquidity reinforce one another, and to fragmented-market research in which liquidity supply determines the trading opportunities available to investors (Krugman, 1980; Chen and Duffie, 2021; Lehar and Parlour, 2025). A vehicle currency is made when it wins existing trades and when it becomes the liquid bridge inherited by new ones.

2 Setting and data

2.1 Routes and notation

Decentralised exchanges distribute liquidity across smart-contract pools that routers can combine atomically. A trader may use one pool, divide an order across several pools, or trade through an intermediary asset. Because the pool calls share a transaction, the intermediary sequence is observable.[2] A route through an intermediary is unavoidable when no direct pool connects the assets; when direct and indirect pools coexist, the router compares their fees, price impact, gas, and venue access. Lehar and Parlour (2025) describe the constant-product design and the economics of liquidity supply in these markets.

Blockchain logs record each pool call in execution order. We recover a route by following the directed token flows within a transaction. Calls with a common source and destination form a split order; a token received in one call and spent in a later call joins the calls into a sequential route. The same rule separates unrelated conversions bundled in one transaction. For a reconstructed sequence $A \to B \to C$, (A, C) is the pair, $A \to B$ and $B \to C$ are its legs, and the full sequence is the route. All three objects are directed by token flow. Each leg executes in a liquidity pool; a pair need not have a direct pool. The observed route endpoints belong to the connected pool component and need not coincide with the endpoints of a broader user instruction.

A transaction on January 10, 2026 illustrates both the information in the logs and its limit. A public explorer describes a 460,000 PYUSD–USDe swap executed through 0x.[3] The user's instruction begins with PYUSD, but the connected pool sequence begins when 459,929.81 USDC enters Fluid. Fluid returns 460,331.05 USDT, which then enters Uniswap v4 and produces 460,104.28 USDe. The observed pair is therefore USDC $\to$ USDe; its legs are USDC $\to$ USDT and USDT $\to$ USDe, and USDT is the vehicle. The earlier PYUSD $\to$ USDC conversion occurs outside the connected pool sequence, so the data do not show whether USDC was another vehicle or inventory supplied by the executor. We apply the connected-component rule to every transaction and do not infer an unobserved leg from nearby transfers. This choice may understate the intermediary role of assets such as USDC when a broader instruction is funded outside the observed pool sequence.

[2] Ethereum records a transaction's ordered logs in its receipt; see the official Ethereum JSON-RPC documentation.

[3] Transaction `0xbda1d07f...9bf67ad9`.

The sequence directly identifies intermediary use. Conditional on reconstructed pool states immediately before execution, it also defines a same-size cost comparison between the observed route and feasible alternative paths. We introduce notation for the route here and develop the cost comparison in Section 4.

Write $\mathcal{A}_t$ for the assets tradable on day t and $\mathcal{P}_t$ for the pools live on that day. For a route r, let $e(r)$ record its block, transaction position, and first pool-event position. The notation $e(r)^-$ refers to the state immediately before that transaction begins; a daily closing state generally occurs later. A route carries an ordered pool sequence $(p_1, \ldots, p_{m(r)})$ and token sequence $(k_0, \ldots, k_{m(r)})$, where $k_0 = i$ is the input and $k_{m(r)} = o$ is the output. Thus (i, o) is the pair and each directed transition $k_{h-1} \to k_h$ is a leg. The dollar input notional q corresponds to $\Delta_i(q, e) = q/P_{i,e}$ units of the input token. Let χ_{p,e^-} denote pool p's state immediately before execution and let $Q_p(i \to j, \Delta_i; \chi_{p,e^-})$ denote the units of token j returned for an exact input of Δ_i units of token i. Composing these pool quotes gives the route's token output and dollar value,

$$\begin{aligned} \Delta_{k_h} &= Q_{p_h}\left(k_{h-1} \to k_h, \Delta_{k_{h-1}}; \chi_{p_h, e(r)^-}\right), & h &= 1, \ldots, m(r), \\ Q_r(q, e(r)) &= \Delta_o, & O_r(q, e(r)) &= P_{o,e(r)} Q_r(q, e(r)). \end{aligned} \tag{1}$$

The venue-specific forms of Q_p appear in Appendix B. Pool invariants take token quantities as arguments, which is why the dollar notional q and the token input Δ_i remain distinct. Equation (1) describes an executed route and, once market state has been reconstructed, prices a counterfactual path.

Intermediary use is one of several currency roles. Invoicing concerns the currency named in trade contracts (Gopinath et al., 2020; Amiti et al., 2022; Mukhin, 2022); funding concerns the denomination of liabilities (Gopinath and Stein, 2021; Eren and Malamud, 2022); and dealer turnover records trading activity (Somogyi, 2026). These roles can reinforce one another. The route in Equation (1) identifies the asset used between the currency sold and the currency bought.

This route setting makes the classical volume–cost mechanism observable at transaction level. Greater use of a vehicle expands trading in the pools supporting its two legs; deeper pools can then lower the cost of using the same vehicle again (Krugman, 1980). In Kiyotaki and Wright (1989), traders accept an asset for indirect exchange when its marketability compensates for its storage cost. Dowd and Greenaway (1993) emphasise the fixed costs of learning, accounting, and quoting in a new currency. Liquidity placement, venue access, and the routes available to software provide corresponding coordination forces on decentralised exchanges.

2.2 Measuring vehicle dominance

An exact two-leg route consists of one pool trade on either side of a unique intermediary. This gives the primary measure a simple unit: each endpoint exchange contributes one intermediary observation. A supplementary multiple-intermediary measure includes routes of any length and counts each intermediary use separately. For route r, let $\mathcal{M}(r)$ denote the set of assets between its source and destination. We record the intermediary status of asset k as

$$Z_{r,k} \;=\; \mathbf{1}\{k \in \mathcal{M}(r)\}.$$

Let $a(k)$ assign each token to an economic asset type. The native group consolidates ETH and its one-for-one wrapped representation, WETH, after route reconstruction. The stable group contains fiat-reserve, on-chain-collateralised, synthetic, and non-dollar stable currencies. For $g \in \{\text{native}, \text{stable}\}$, define $Z_{r,g} = \sum_{k:a(k)=g} Z_{r,k}$; on an exact two-leg route, this sum is an indicator.

This grouping preserves a token-level estimand. WETH is the one-for-one token representation that allows ETH to interact with ERC-20 pool contracts; wrapping changes the technical form, not the underlying settlement exposure. We therefore reconstruct routes at the token level and consolidate ETH and WETH only when assigning the economic asset used as a vehicle. Dollar-pegged tokens remain distinct instruments because their issuers, redemption arrangements, risks, and leg-level liquidity differ. USDT in the USDC–USDT–USDe example is therefore an intermediary token even though all three instruments target the same unit of account. The aggregate stable group measures dominance by a family of stable tokens, while the token-level results retain issuer-specific competition.

Native assets and stablecoins define the main comparison because they are the two persistent broad vehicle families in the route panel. Staked-native, imported, and other intermediaries remain visible in the wider annual series and supplementary results; they do not enter the native-versus-stable denominator. For routes with more than one intermediary, realised intermediary counts retain each asset used along the route. Graph betweenness is useful for describing how often an asset could lie on feasible multi-leg paths. It measures the opportunity set, while the dominance measure records the routes traders actually use.

We begin with the competition between native and stable intermediaries. Let $\mathcal{R}_t^{(2),NS}$ denote exact two-leg routes on day t whose intermediary belongs to either group, and let $\mathcal{R}_t^{(2),NS,V}$ retain routes for which the source, intermediary, and destination values agree within 20%. The equally weighted and dollar-weighted group shares are

$$S_{g,t}^{N} = \frac{\sum_{r\in\mathcal{R}_t^{(2),NS}} Z_{r,g}}{|\mathcal{R}_t^{(2),NS}|}, \qquad S_{g,t}^{V} = \frac{\sum_{r\in\mathcal{R}_t^{(2),NS,V}} \nu_r Z_{r,g}}{\sum_{r\in\mathcal{R}_t^{(2),NS,V}} \nu_r}, \tag{2}$$

where ν_r is the arithmetic mean of the route's total dollar value at its observed source and destination. The 20% rule also requires every intermediary's incoming and outgoing dollar values to have a ratio between 0.8 and 1.2. Thus $S_{\text{stable},t}^{N}$ is stablecoins' aggregate share among native and stable intermediaries, and $S_{\text{stable},t}^{V}$ is the corresponding dollar-weighted share. The two groups exhaust this denominator. Share levels are reported in percent; the unit for absolute share changes is one percentage point (pp).

The frequency with which an asset appears at the beginning or end of an observed pool route provides a benchmark for its intermediary use. This comparison uses a broader set of currencies $\mathcal{C}$: native, staked-native, stable, and imported currencies, with unclassified assets excluded. We retain every complete route that does not return to its starting currency. Direct routes contribute to the source and destination counts, and a longer route contributes one observation for each intermediary it uses. Let $n^I_{k,t}$ count intermediary uses and $n^E_{k,t}$ count source or destination appearances for $k \in \mathcal{C}$. The two shares are

$$I^N_{k,t} = \frac{n^I_{k,t}}{\sum_{j\in\mathcal{C}} n^I_{j,t}}, \qquad E^N_{k,t} = \frac{n^E_{k,t}}{\sum_{j\in\mathcal{C}} n^E_{j,t}},$$

and the corresponding share gap and excess-use ratio are

$$\Delta^N_{k,t} = I^N_{k,t} - E^N_{k,t}, \qquad X^N_{k,t} = \frac{I^N_{k,t}}{E^N_{k,t}} \quad \text{when } E^N_{k,t} > 0. \tag{3}$$

In Equation (3), a positive $\Delta^N_{k,t}$ means that the asset's share of intermediary use exceeds its share of source and destination appearances. We interpret $X^N_{k,t}$ as intermediation intensity relative to endpoint demand: zero means the asset is not used as an intermediary, one means its two shares are equal, and a value above one means it is overrepresented as a vehicle. The ratio is not a ranking of aggregate vehicle dominance. Dollar-weighted measures replace counts with route values while retaining the same currencies and the 20% agreement rule. Because this role-specialisation measure compares each currency's intermediary and endpoint uses, its denominator differs from the native-or-stable denominator in Equation (2). The latter remains the measure of aggregate rotation.

Source and destination assets must differ in the analysis sample because vehicle exchange connects two distinct endpoint assets. We classify routes that return to their starting asset as round trips and remove them. Such routes may contain cyclic arbitrage or other self-returning strategies: their economic object is a trading cycle, not an exchange from one endpoint asset into another, although trader intent remains unobserved. Daian et al. (2020) show that several arbitrage trades can be bundled atomically. On the median of 79 sampled days, round trips account for 12.7% of multi-leg routes by count and 21.7% by value. Appendix A reports their distribution across the sampled dates.

Within this route sample, we classify assets by their economic exposure: native, staked native, stable, imported, and other. The native category contains the platform's settlement asset. We treat ETH and its one-for-one wrapped representation, WETH, as one economic asset after reconstructing the route. This consolidation is an economic classification, not an address-level identity (Milionis et al., 2022). Liquid-staking tokens remain separate because they also convey staking returns. Stable assets target a fiat value, imported assets are wrapped representations of off-chain or cross-chain value, and the remaining assets enter the residual category; the asset's economic exposure determines its category. Stable assets themselves encompass economically different designs. Reserve-backed stablecoins rely on backing and redemption, making confidence in those arrangements relevant to acceptance and run risk (Gorton and Zhang, 2023); fiat-backed, crypto-collateralized, and algorithmic stablecoins also differ in their solvency and peg-restoration mechanisms (Catalini et al., 2022; Lyons and Viswanath-Natraj, 2023). Appendix G examines the two largest reserve-backed stablecoins separately.

2.3 The route panel

The principal route panel covers Curve, Uniswap v2, v3 and v4, Balancer, SushiSwap v2 and v3, and Fluid on Ethereum. Their indexed pool records or factory registries link each pool to its token contracts. Uniswap v1 enters the separate protocol-architecture sample in Section 5.2. The original v1 query retained `exchangeAddress` but did not request `tokenAddress`; no local file supplies the missing exchange-to-token crosswalk. Re-fetching the v1 exchange entity with its token address and symbol would add those identities. Shared transaction hashes and matching ETH legs already recover forced token-to-token routes, so the architecture result does not require that additional fetch. Table 1 summarises the principal panel's coverage. The sample spans 2,332 dates from February 11, 2020 through June 30, 2026. These deployments include several major exchanges but remain a subset of Ethereum trading. Protocols enter after their launch dates, so the exchanges and routes available to traders expand over the sample. We keep 55 early dates with no observed swaps as zero-activity dates for the covered deployments.

We begin with 472,254,909 pool-level swap records. Indexed blockchain data supply events for seven exchange series, Dune's `dex.trades` table supplies Fluid records, and direct Ethereum queries supply block headers, ordered logs and receipts, token precision, factory registries, and state checks.[4] After harmonising trade direction, event identity, and token units, the panel contains 471,616,269 legs. We order those legs within each transaction and apply the flow rule above to distinguish single-pool trades, split orders, sequential routes, and unrelated calls. Appendix A gives the reconstruction algorithm and reports the historical token units that we cannot verify.

Transactions with more than one disconnected pool-event group pose a separate measurement limitation: the observed calls do not reveal whether the groups belong to one broader instruction (Xi and Moallemi, 2026). The principal sample excludes every component in those transactions. The excluded share rises from 0.8% of reconstructed components in 2024 to 6.2% in 2026. Appendix Table 7 treats each internally valid component as a separate route. Under that construction, stablecoins' count share rises from 16.9% in 2024 to 43.7% in 2026, and their value share rises from 32.9% to 77.3%. The stricter connected-transaction rule therefore understates the measured rotation slightly.

Every token sequence retained by the connected-transaction rule enters the count measure. Dollar-weighted results require source, intermediary, and destination values to agree within 20%. This is an eligibility test for a coherent dollar notional, not a control for route characteristics: covariates cannot repair a route whose token values do not reconcile. Keeping the count sample broader preserves observed intermediary choice when valuation is noisy; each value exhibit reports the stricter sample's coverage.

Transactions and contract-state changes are publicly observable (Schär, 2021). Reconstructing a route still requires matching pool events, venues, and token units. This procedure identifies the executed route and its calling contract, but an executor address does not establish which frontend or

[4]See The Graph's indexing documentation, Dune's curated-data documentation, and the official Ethereum node-interface documentation.

software formed the quote. Existing router labels likewise cover only part of decentralised-exchange activity (Xi and Moallemi, 2026). On a 2024 snapshot of 74,323 swaps, 241 distinct senders appear, and a registry of the ten largest accounts for 67.7% of those swaps. By a 2025 snapshot, 397 senders appear across 130,282 swaps and the same registry covers 11.8%. We therefore report route-level results without attributing them to a particular routing service.

Table 1: The route panel

	Route panel
Dates in the full route panel	2,332
Missing source-days	0
Raw swap rows	472,254,909
Usable directed route legs	471,616,269
Routes per day, median of 79 sampled days	161,152
Multi-leg routes per day, median of 79 sampled days	31,253
Multi-leg share of routes, median of 79 sampled days	19.7%
Round trips, share of multi-leg routes by count	12.7%
Round trips, share of multi-leg routes by value	21.7%

Note: The full eight-deployment panel contains UTC dates from 11 February 2020 through 30 June 2026. Raw swaps and usable legs use that full panel. The separate Uniswap v1 architecture sample lies outside these totals because its retained events do not recover pair token identities uniformly. Daily route statistics and round-trip shares are validation quantities computed on 79 dates sampled across 1 June 2020 through 27 April 2026; this computational sample is not the support of the main dominance estimates. A multi-trade route traverses more than one pool; a round trip ends in the token with which it began. Round-trip shares are formed among multi-trade routes.

3 How stablecoins gain the vehicle role

3.1 The aggregate rotation

Stablecoins carry a much larger share of vehicle activity in 2026 than in 2024. The comparison uses the same January–June dates in both years because the 2026 panel ends on June 30. Stablecoins' share among native and stable intermediaries rises from 16.9% to 42.3% by route count and from 32.7% to 76.5% by routed value. Table 2 reports the paired-day estimates.

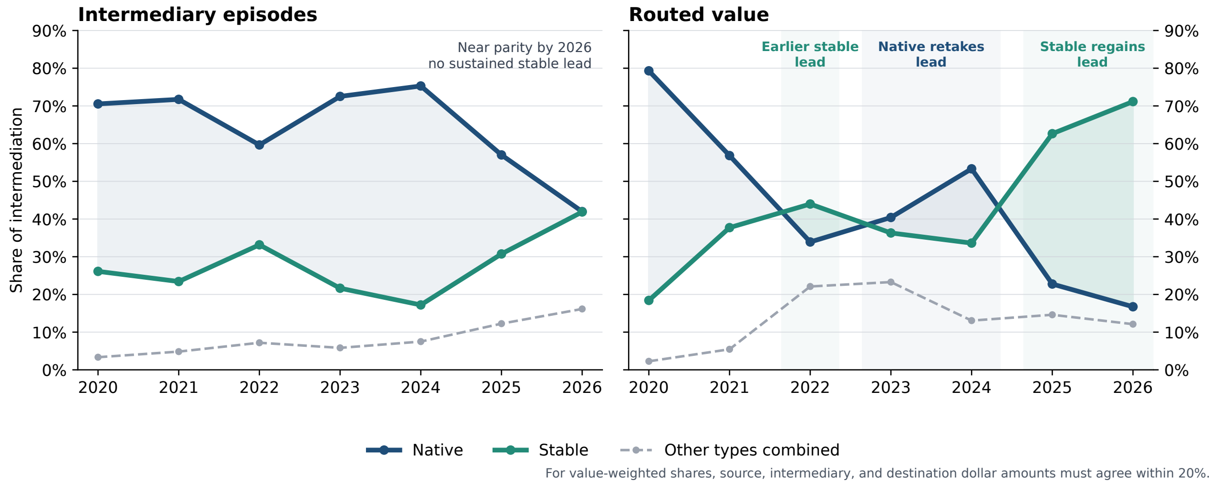
The annual series in Figure 1 shows that the change is not monotonic. Native assets regain share in 2023 and 2024; stablecoins recover the routed-value lead in 2025. The 20% value-agreement

Table 2: Stablecoin share among native and stable intermediaries

Stablecoin share among native and stable intermediaries	2024	2026	Change [pp] (s.e.)
Route count	16.9%	42.3%	+25.4 (1.05)
Dollar-weighted routes (20% agreement)	32.7%	76.5%	+43.9 (2.02)

Note: The denominator contains exact two-leg routes whose sole intermediary is native or stable. Daily shares receive equal weight over matched January–June dates in 2024 and 2026. Dollar-weighted estimates require source, intermediary, and destination values to agree within 20%. Standard errors use a Newey–West covariance with a 30-day Bartlett bandwidth.

Figure 1: Intermediary activity rotates between native assets and stablecoins



Note: The sample contains complete routes that do not return to their starting asset. Each route contributes once for every intermediary it uses, so routes with more than two legs can contribute several observations. Shares are formed across all intermediary types; Table 2 uses the narrower native-or-stable denominator. The 2026 observation covers January through June.

sample covers 71.6% of native intermediary value and 55.1% of stablecoin intermediary value in 2026. Fiat-reserve tokens account for +24.50 pp of the +25.42 pp count increase and +44.09 pp of the +43.87 pp value increase. The aggregate movement is therefore mainly a shift toward reserve-backed dollar tokens, although the token-level results below show that USDT and USDC do not move together.

3.2 Pair formation and persistence

Aggregate dominance can change inside continuing pairs or through the distribution of activity across pairs. Let p denote a pair and $y \in \{2024, 2026\}$. A continuing pair carries native-or-stable routes in both matched periods; a year-specific pair carries them in only one. Let W_y be the share of native-plus-stable activity in continuing pairs, $\omega_{p,y}$ pair p's weight within that activity, and $s_{p,y}$ its stablecoin share. Continuing pairs have stablecoin share $S_{C,y} = \sum_p \omega_{p,y} s_{p,y}$, while year-specific pairs have share $S_{E,y}$. Writing $\Delta x = x_{2026} - x_{2024}$ and $\bar{x} = (x_{2024} + x_{2026})/2$ gives the exact midpoint decomposition

$$\Delta S = \underbrace{\bar{W} \sum_p \bar{\omega}_p \Delta s_p}_{\text{switching within continuing pairs}} + \underbrace{\bar{W} \sum_p \bar{s}_p \Delta \omega_p}_{\text{activity reallocation}} + \underbrace{(\bar{S}_C - \bar{S}_E)\Delta W}_{\text{continuing-pair weight}} + \underbrace{(1-\bar{W})\Delta S_E}_{\text{year-specific pairs}} . \tag{4}$$

We also compare the same pair, date within the year, and realised single- or cross-exchange route class. For route-count or routed-value stablecoin share $S^{(m)}_{pds,y}$, the matched regression is

$$S^{(m)}_{pds,y} = \alpha^{(m)}_{pds} + \beta^{(m)}\mathbf{1}\{y = 2026\} + \varepsilon^{(m)}_{pds,y}, \qquad m \in \{N, V\}. \tag{5}$$

Here d is the month-day and s the realised route class. Observations are weighted by native-plus-stable route count or supported routed value; standard errors cluster by pair and date. The comparison is descriptive because route class and trading activity can respond to the same forces as vehicle choice.

Table 3 applies Equation (4) to route counts and supported value and reports Equation (5) on cells defined by pair, calendar date, and route class.

The within-pair component is -0.1 pp by route count and 0.0 pp by routed value. These net estimates combine $+1.3$ pp from pairs moving toward stablecoins and -1.4 pp from pairs moving toward the native asset. Activity reallocation across continuing pairs contributes $+8.6$ pp by count and $+26.2$ pp by value, while year-specific pairs contribute $+17.8$ pp and $+19.2$ pp. Equation (5) reaches the same conclusion on the more restrictive matched sample: stablecoin share changes by $+0.2$ pp by count and -1.3 pp by value. A large aggregate rotation therefore coexists with little net switching inside established cells defined by pair, calendar date, and route class.

Endpoint demand sharpens the composition result. A pair with WETH at one endpoint cannot also use WETH as its intermediary, so the native-versus-stable choice is mechanically fixed. Such pairs supply $+6.3$ pp of the $+8.6$ pp count reallocation and $+13.9$ pp of the $+21.0$ pp entry component. An exact decomposition of the headline daily change by endpoint direction shows that pairs with one native and one stable endpoint contribute $+9.2$ pp of the $+25.4$ pp count increase and $+10.1$ pp of the $+43.9$ pp value increase. Pairs with two stable endpoints contribute only $+1.0$ pp by count but $+13.2$ pp by value. Within that value channel, USDT alone contributes $+13.7$ pp; USDC, DAI, and the remaining stablecoin intermediaries together contribute -0.5 pp. The stable-to-stable result is present on all 181 common days, and trimming the largest decile of daily contributions leaves a $+12.3$ pp increase. The largest component still comes from all other pairs, at $+15.2$ pp by count and $+20.6$ pp by value. Stablecoin dominance therefore spreads beyond pairs whose endpoint identity limits the native vehicle, while routing between stable currencies concentrates on one intermediary issuer.

The first routes in a new pair predict its subsequent vehicle. For entrant pair p first observed on date t, let $S^{\text{entry}}_{p,t}$ be its initial stablecoin share and $S_{p,t+h}$ its stablecoin share through horizon h. We estimate

$$S_{p,t+h} = \alpha_h + \beta_h S^{\text{entry}}_{p,t} + \gamma_h \mathbf{1}\{S^{\text{entry}}_{p,t} > 1/2\} + X'_{p,t}\delta_h + \varepsilon_{p,t+h}, \tag{6}$$

where $X_{p,t}$ contains entry size, cohort year, endpoint class, direct-route share, and route-complexity share. The sample excludes pairs with WETH at an endpoint and requires complete follow-up support.

Table 3: Decomposition of the stablecoin rotation

Component or estimate	Estimate [pp]	Obs.
Panel A. Route-count share: market activity and vehicle incidence		
Pairs entering or leaving the sample	+9.8	
Pairs gaining or losing a vehicle route	−0.4	
Market activity shifting across continuing pairs	+7.9	
Change in how often continuing pairs use a vehicle	+7.0	
Stablecoin share within continuing vehicle-using pairs	+1.3	
Total route-count change	+25.7	
Panel B. Route-count share: continuing and year-specific pairs		
Net stablecoin-share change within continuing pairs	−0.1	
Pairs moving toward stablecoins (1,575)	+1.3	
Pairs moving toward native assets (1,489)	−1.4	
Vehicle activity shifting across continuing pairs	+8.6	
Weight of continuing versus year-specific pairs	−0.5	
Pairs traded in only one year	+17.8	
Total route-count change	+25.7	
Panel C. Dollar-weighted share: continuing and year-specific pairs		
Net stablecoin-share change within continuing pairs	0.0	
Pairs moving toward stablecoins (1,511)	+2.3	
Pairs moving toward native assets (1,445)	−2.4	
Vehicle activity shifting across continuing pairs	+26.2	
Weight of continuing versus year-specific pairs	−2.5	
Pairs traded in only one year	+19.2	
Total change in dollar-weighted share	+42.8	
Panel D. Matched pair estimates		
All two-leg routes, count share	+0.22 (0.76)	188,520
20% agreement sample, count share	+0.32 (0.75)	182,834
20% agreement sample, dollar-weighted share	−1.35 (2.19)	182,834

Note: Panels A through C compare pooled activity over matched January–June dates in 2024 and 2026. Panels B and C report Equation (4); their indented rows split the within-pair component into positive and negative contributions. Panel C weights routes by dollar value and applies the 20% agreement rule. Panel D estimates Equation (5); standard errors use two-way clustering by pair and date. Panels A and B factor the same route-count change in different ways, so their components should not be combined.

Table 4 shows that a 1 pp higher initial stablecoin share predicts +0.86 pp higher stablecoin share over 30 days and +0.74 pp over 120 days. The 120-day coefficient is +0.89 pp among pairs with no direct route at entry and +0.78 pp when routes are weighted by supported value. Persistence extends to issuer identity: a 1 pp higher entry share for a named stablecoin predicts +0.97 pp higher own-token share over 120 days. These estimates describe persistence from initial vehicle use; the initial vehicle is not randomly assigned.

Network reach, issuer-level specialisation, within-day endpoint demand, and exchange span provide supplementary views of the same formation result. Appendix G reports these estimates. They show that stablecoin arrival is more common in older and more active pairs, intermediary use moves closely with a currency's endpoint demand within a day, and the stablecoin rotation appears in both single- and cross-exchange routes. These comparisons help locate the result but do not replace the direct decomposition in Table 3.

Table 4: Pair entry and later vehicle use

Model	Outcome	Regressor	Coefficient / (s.e.)	Obs. / clusters
Entry persistence	Stable follow-up share ($S_{p,t+h}$), 30 days	Entry stable share ($S^{\text{entry}}_{p,t}$)	$+0.86^{***}$ (0.06)	599,870 / 333
Entry persistence	Stable follow-up share ($S_{p,t+h}$), 120 days	Entry stable share ($S^{\text{entry}}_{p,t}$)	$+0.74^{***}$ (0.13)	512,201 / 243
Entry persistence	Stable-majority follow-up, 120 days	Stable-majority entry ($\mathbf{1}\{S^{\text{entry}}_{p,t} > 1/2\}$)	$+57.3^{***}$ (14.8)	512,201 / 243
No-direct-route split	Stable follow-up share ($S_{p,t+h}$), 120 days	Entry stable share ($S^{\text{entry}}_{p,t}$), no direct route	$+0.89^{***}$ (0.13)	512,020 / 243
Value-weighted entry	Stable value share, 120 days	Entry stable value share	$+0.78^{***}$ (0.14)	427,625 / 243
Named-stable identity	Own-stable follow-up share, 120 days	Entry named-stable share	$+0.97^{***}$ (0.02)	6,087 / 243
Secure-volume endpoint	Entry stable share ($S^{\text{entry}}_{p,t}$)	Stable endpoint × 2026	$+5.7^{***}$ (1.3)	615,333 / 363
Route architecture	Entry stable share ($S^{\text{entry}}_{p,t}$)	Direct-route share ($D_{p,t}$) × 2026	$+0.50^{***}$ (0.14)	615,333 / 363
Route architecture	Entry stable share ($S^{\text{entry}}_{p,t}$)	Complex-route share ($C_{p,t}$) × 2026	$+0.36^{***}$ (0.12)	615,333 / 363

Note: The entry-persistence rows estimate Equation (6). Coefficients appear above entry-date-clustered CR1 standard errors in parentheses. Share outcomes and share regressors are measured in percentage points (pp). The unit is an entrant pair by route class without WETH at either endpoint, except for the named-stable identity row, whose unit is a stablecoin entry observation with asset fixed effects. Rows are route-weighted except the stable-value row, which uses routed value satisfying the 20% agreement rule. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

4 Comparing direct and intermediated execution

Realised routes identify the assets that carried each trade. Execution cost is a separate object: it compares those routes with alternatives available immediately beforehand, quoted at the same trade size and over a common set of exchanges. We hold the source i, destination o, dollar input q, and pre-transaction pool state e^- fixed. The best direct and two-leg outputs through intermediary k are

$$O^D_{i,o,q,e} = P_{o,e} \max_{p \in \mathcal{P}_{i,o,e^-}} Q_p\big(i \to o, \Delta_i(q,e); \chi_{p,e^-}\big),$$

$$O^I_{i,o,k,q,e} = P_{o,e} \max_{p' \in \mathcal{P}_{k,o,e^-}} Q_{p'}\left(k \to o, \max_{p \in \mathcal{P}_{i,k,e^-}} Q_p\big(i \to k, \Delta_i(q,e); \chi_{p,e^-}\big); \chi_{p',e^-}\right)$$

This comparison extends the realised-versus-optimised allocation benchmark of Xi and Moallemi (2026) from parallel pools connecting the same two assets to sequential routes through an intermediary.

The vehicle-route shortfall relative to direct execution is

$$\Delta C^D_{i,o,k,q,e} = \frac{O^D_{i,o,q,e} - O^I_{i,o,k,q,e}}{O^D_{i,o,q,e}},$$

reported in basis points as $10^4 \Delta C^D_{i,o,k,q,e}$. Positive values mean that the direct route returns more output before gas. Swaps and liquidity changes are replayed in on-chain order because a daily or hourly closing state incorporates events that occurred after the trade. The observed event marks settlement; a quote formed earlier can reflect a different opportunity set.

We compare hypothetical legs only within a common executable range. Let $\widetilde{Q}_\ell$ denote leg ℓ's output at the pool's marginal rate. Each leg satisfies

$$\pi_\ell(\Delta_i, e^-) = \frac{\widetilde{Q}_\ell(\Delta_i, e^-) - Q_\ell(\Delta_i, e^-)}{\widetilde{Q}_\ell(\Delta_i, e^-)} \leq \bar{\pi}, \qquad \bar{\pi} = 0.05. \tag{7}$$

The 5% own-price-impact ceiling applies across pool families; contract-specific limits add further restrictions. Appendix B gives the pricing equations and Appendix D calibrates the support rule.

Executable price differences can persist when trading is segmented across cryptocurrency markets (Makarov and Schoar, 2020). Prices therefore offer a demanding rival explanation for persistent vehicle use. If the intermediary merely records the price in one locally attractive pool, widening the same-state opportunity set should often change the intermediary. We test that prediction on the fifteenth day of every month from June 2020 through June 2026. Each observed two-leg route must reproduce realised output within one basis point. We then widen the opportunity set in three stages: alternative paths using the same vehicle and venue families as the observed route, paths using the same vehicle across Uniswap v2, SushiSwap v2, and Uniswap v3, and paths using any named public vehicle or the direct connection across those three exchanges. The main sample also requires coherent leg values, at least \$100 of input, and the 5% price-impact bound on the realised route.

Table 5 shows that the chosen vehicle survives broader price competition. Only 6.6% of routes have a same-vehicle alternative that improves gross output by more than one basis point within the venue families already used. Opening all three exact venues raises that share to 44.4%. Opening the vehicle set and the direct path raises it only another 2.0 pp, to 46.4%. Among routes with an improvement, the median gain is 21.9 bp. Thus public venue prices often favour a different venue while preserving the intermediary asset.

The vehicle-share comparison gives the same result. The full price set lowers stablecoin vehicle share by 1.2 pp on a route-weighted basis and 2.1 pp when routes are weighted by input value. The route-weighted standard error is 0.2 pp. Restricting the comparison to the 78,637 routes with at least \$10,000 of input lowers the route-weighted share by 1.8 pp. Vehicle identity is therefore much less sensitive to broader public prices than venue choice. This separation supports an asset-level coordination role alongside venue-level execution frictions; it does not identify why a router omitted an available pool.

Receipt gas measures the fixed execution toll attached to adding a vehicle leg. That toll matters because gas fees can change both trading costs and liquidity concentration in decentralised exchange (Caparros et al., 2024). In the executed-route gas sample, a stable-vehicle route in 2024 uses 94,290 more gas than the median direct route, 50.2% of the direct-route median. Evaluated at that year's median gas price and WETH price, the extra toll is \$3.24, so a trade needs about \$32.4k of notional before the extra gas is below one basis point. In 2026 the median stable-vehicle route uses 233,640 more gas than the direct-route median, but the dollar toll falls to \$0.08 and the one-basis-point notional falls to \$762. Pricing the same fixed toll at each stable-vehicle route's own gas

Table 5: Exact pre-transaction prices and vehicle identity

Panel A. Coverage and quote validation				
Linear routes in the exact venue set [%]				93.4
Mapped observed routes reproduced within 1 bp [%]				89.1
Panel B. Nested opportunity set				
	Realised route	Same vehicle, used venues	Same vehicle, all exact venues	Any named vehicle or direct
Routes with more than 1 bp higher output [%]	0.0	6.6	44.4	46.4
Stablecoin vehicle share [%]	19.5	19.5	19.5	18.4
Full-set minus realised stablecoin share [pp]			Route weighted	-1.16^{***} (0.19)
			Input-value weighted	-2.09^{***} (0.31)
Routes / dates				777,651 / 73

Note: The sample contains exact two-leg routes executed on Uniswap v2, SushiSwap v2, or Uniswap v3 on the fifteenth day of each month from June 2020 through June 2026. Source, destination, input amount, and pre-transaction pool state remain fixed. Each observed quote reproduces realised output within one basis point. Leg values must agree within 20%, input must be at least $100, and every observed or hypothetical leg must remain below 5% own-price impact. The public vehicle set contains WETH, USDC, USDT, DAI, and WBTC; a realised vehicle outside that set remains eligible. A route changes vehicle only when gross output improves by more than one basis point. Standard errors in parentheses are clustered by date. P-values use Holm adjustment across route- and input-value-weighted comparisons in five reported samples. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. The three exact venues contain 93.4% of linear routes over the sampled dates, and 89.1% of mapped observed routes reproduce within one basis point. Coverage falls as other exchanges expand late in the sample. The comparison is gross of gas and excludes private order flow. 98 routes imply more than twice realised output under standard invariant pricing; removing them leaves every displayed share unchanged at one decimal place, and they are excluded from the median-gain calculation.

price sharpens the interpretation: median stable-vehicle notional falls from \$832 to \$82, while the median fixed toll falls from 44.3 bp to 16.2 bp and the share of stable-vehicle routes with a fixed toll below 25 basis points rises from 39.5% to 55.7%, the trade-size version of the fixed-cost channel in Caparros et al. (2024). Thus the extra-hop overhead remains visible in gas units while becoming economically small for ordinary routed notionals. This evidence supports the interpretation that vehicle formation spread to routes for which fixed execution overhead had ceased to be a large barrier. The estimand is the executed route's fixed gas toll; a stable-versus-WETH all-in price ranking remains outside this check.

The benchmark represents public on-chain liquidity in the declared exchange set. Private order flow, financing costs, losses from reverted transactions, and pools outside that set can alter a router's choice. Pool-family coverage is economically material. In the sampled Curve exercise, pools outside the StableSwap model carry 34.2% of assessed volume, including 65.2% of native-leg volume and 21.1% of stablecoin-leg volume. Hook-bearing or dynamic-fee Uniswap v4 pools, SushiSwap v3, and several Balancer families also use pricing states outside the benchmark. Appendix E quantifies these coverage limits.

5 How vehicle use becomes established

The pair decomposition locates the aggregate rotation in the pairs that become active and attract trading. The next question is how a vehicle becomes available inside those pairs and why its use persists. Longer routes carry the same rotation. The transition from Uniswap v1 to v2 separates inherited route structure from voluntary vehicle choice. Realised vehicle use then links the aggregate result to the capital on the two legs required to complete the route.

5.1 The rotation spans short and long routes

Every complete route enters Figure 1. A route with several intermediary assets contributes one observation for each intermediary position, so the shares divide the work performed along the full route. Table 2 uses exact two-leg routes for a narrower reason: one route then has one intermediary and maps to one vehicle choice.

Stablecoins gain across both short and long routes. Among two-leg routes, stablecoin share rises by +25.4 pp by count and +43.9 pp by supported value between matched dates in 2024 and 2026. Among intermediary positions on routes with more than two legs, it rises by +16.3 pp and +35.9 pp. Splitting both groups again by whether all legs occur on one exchange leaves four cells, and all four show a stablecoin increase. The smallest increases are +17.0 pp by count and +25.6 pp by value. The result therefore appears in the full route network as well as in the one-intermediary choice sample.

Appendix Table 15 reports two all-route normalisations. Intermediary-position share divides a currency's uses by all intermediary positions and sums to one across currencies. Route-participation share divides routes containing the currency by all routes and can sum above one because a longer route can use several currencies. Stablecoin participation rises from 17.6% in 2024 to 46.1% in 2026 H1.

The same table compares realised use with betweenness in the observed trading graph. WETH ranks first in all seven annual graphs and retains betweenness of 0.927 in 2026. On the monthly graph dates, however, its intermediary-position share falls from 76.2% in 2024 to 42.3% in 2026, while USDC and USDT together rise from 14.9% to 37.4%. Binary network position therefore changes much less than realised intermediary use. An edge records that at least one leg connects the two assets; local depth and execution cost determine how useful that connection is for a particular route.

5.2 Protocol architecture and inherited vehicle use

Uniswap v1 created pools between an ERC-20 token and ETH, so token-to-token trades crossed ETH. Uniswap v2 allowed arbitrary ERC-20–ERC-20 pools and made direct token-to-token trading possible.[5] Forced ETH intermediation accounts for 8.0% of Uniswap v1 swap transactions and 8.1% of its ETH volume over the venue's life, and about one trade in ten in 2019 and 2020. One transaction from March 2020 makes the route visible: a trader selling 250 MKR received 439.13 ETH at one exchange contract and spent the identical ETH amount at a second for 49,892.40 DAI.[6]

Native-asset pairing remained pervasive after v2 opened direct token-to-token pools. The share of single-leg Uniswap v2 trades executed in WETH pools was 95.1% in 2020 and 95.5% in 2026. Among the 477,633 token combinations that ever traded on Uniswap v2, 97.1% include WETH, and WETH's share of newly traded token combinations rose from 84.1% in 2020 to 97.9% in 2026. Early v2 pools inherited liquidity, users, and routing conventions from v1, which makes part of the early persistence mechanical. Deep WETH pools and later launch conventions then carried that structure forward. The pair-entry and bridge-depth estimates below examine settings in which stablecoin and WETH routes can coexist.

5.3 Local bridge depth and route allocation

The classical vehicle-currency mechanism links trading volume to lower transaction costs (Krugman, 1980). Here an indirect route needs two legs: one connecting the source to the vehicle and another connecting the vehicle to the destination. Capital on only one leg is insufficient. The weaker leg therefore determines the deposited-capital depth of the bridge. Deposited capital remains an equilibrium stock; a size-specific pretrade quote gives the closer measure of executable liquidity. Concentrated-liquidity providers can also change executable depth by moving capital along the price curve (Adams et al., 2021; Lehar and Parlour, 2025).

We date the first persistent stablecoin bridge for a pair that previously routed through WETH

[5] See Adams, Zinsmeister, and Robinson, *Uniswap v2 Core*, pp. 1–2.

[6] Transaction `0x4dca160d...96a0ca16`, block 9,674,728, March 15, 2020. The transaction is the largest ETH route among the 129,810 mandate-era token-to-token transactions whose two ETH legs match exactly. Public transfer records verify the endpoint tokens.

and had no observed stablecoin route. A bridge begins when DAI, USDC, or USDT has positive prior-calendar deposited capital on both required legs for at least 24 of the next 30 days. Panel A of Table 6 compares the 30 days before support with the first month and with days 30 through 119. Across 865 events, stablecoin route share rises by +7.93 pp in the first month and supported-value share by +3.11 pp. The corresponding count-share increase over days 30 through 119 is 8.68 pp.

The depth comparison follows each bridge after support appears. Let $K_{av,t-1}$ denote prior-calendar deposited capital on the leg joining endpoint a to vehicle v. The stablecoin set is $\mathcal{S} = \{\mathrm{DAI}, \mathrm{USDC}, \mathrm{USDT}\}$. For pair (i, j), define

$$D_{et}^{S} = \max_{v \in \mathcal{S}} \min\{K_{iv,t-1}, K_{vj,t-1}\}, \qquad D_{et}^{W} = \min\{K_{iW,t-1}, K_{Wj,t-1}\},$$
$$Q_{et} = \frac{D_{et}^{S}}{D_{et}^{S} + D_{et}^{W}}, \qquad S_{et}^{R} = \beta Q_{et} + \alpha_e + \lambda_{m(t)} + \eta_{a(t)} + \varepsilon_{et}. \tag{8}$$

Here W denotes WETH, S_{et}^{R} is stablecoin route share, $m(t)$ is calendar month, and $a(t)$ groups days since support into seven-day intervals. Event effects compare changes in relative depth within the same pair and route scope.

Panel B of Table 6 reports the relation. During the first month, the best stablecoin bridge has less than one-tenth of WETH depth on 7,752 of 11,327 comparable pair-days. Stablecoins carry 2.4% of routes in that range. Their route share reaches 53.0% when stablecoin depth matches WETH depth and 69.9% when it reaches twice WETH depth. Within the same bridge event, a 1 pp increase in Q_{et} is associated with +0.64 pp higher stablecoin route share.

Panel C separates support from use. Only 8.3% of the 865 events use one of their supported stablecoins on the support date; 52.8% do so within 30 days and 60.1% within 120 days. First-month use is 42.6% when stablecoin depth is below one-tenth of WETH depth and 84.1% once it reaches that threshold. Controls for prior WETH activity, endpoint direction, route scope, the supported stablecoins, and event month leave a +25.2 pp difference. Among pairs with two non-stablecoin endpoints, the difference is +35.8 pp.

Figure 2 follows the 246 events whose supported stablecoin is first used 7 to 119 days after persistent support. Stablecoin bottleneck capital rises +0.86 log points during the week before use. Relative to the matched WETH bridge, the pre-use increase is +0.78; the pre-minus-post contrast is +1.24. Capital is already present before the first observed stablecoin route. The timing is consistent with traders using the route after it becomes sufficiently deep and with providers anticipating later use. The observed sequence alone does not separate those two directions.

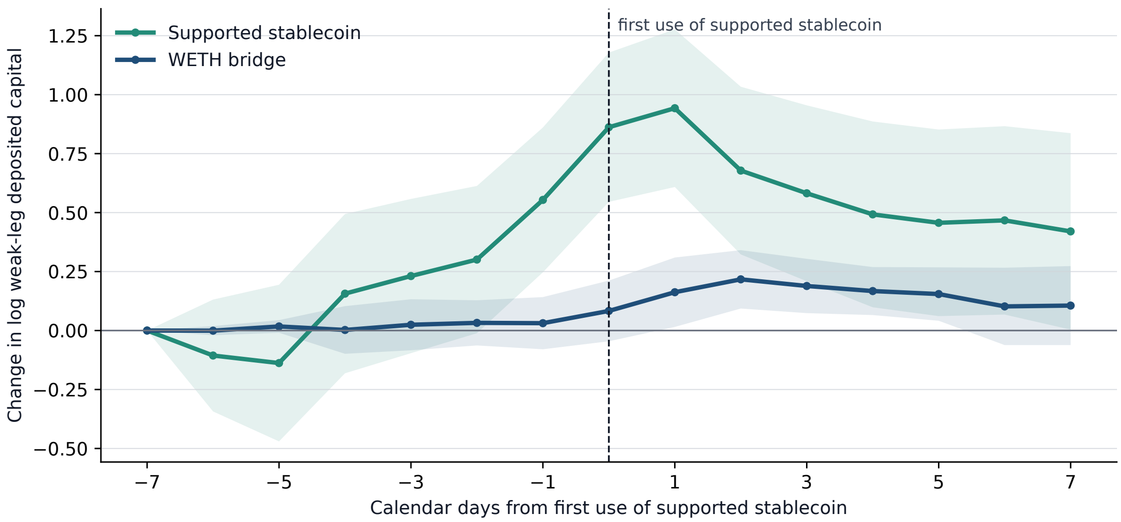
These results connect pair formation to local liquidity. A stablecoin route often becomes technically available while its bridge remains shallow. Use rises sharply as the two-leg bridge approaches WETH depth, and capital builds before first observed use. Pool activation, route-capital feedback, router releases, venue pricing families, and V3/V4 provider activity offer further descriptive evidence. Appendix H reports those comparisons and keeps them separate from the paper's central decomposition and bridge-depth result.

Table 6: Stable-bridge establishment, relative depth, and route allocation

Outcome	First 30 days	Days 30–119
Panel A. Changes around first persistent stable support		
Stable route share [pp]	$+7.93^{***}$ (1.88)	$+8.68^{***}$ (2.57)
Stable supported-value share [pp]	$+3.11^{**}$ (1.53)	$+6.94$ (4.53)
$\log(1 + \text{native routes per active day})$	-0.20^{***} (0.04)	-0.39^{***} (0.06)
$\log(1 + \text{total routes per active day})$	-0.10^{**} (0.04)	-0.29^{***} (0.05)
$\log(1 + \text{native supported value per active day})$	-0.17 (0.11)	-0.61^{***} (0.19)
$\log(1 + \text{total supported value per active day})$	$+0.06$ (0.10)	-0.32^{*} (0.18)
Bridge events	865	818
Panel B. Stable-bridge competitiveness relative to WETH		
Relative-depth slope [pp per +1 pp]	$+0.64^{***}$ (0.07)	$+0.75^{***}$ (0.05)
Stable route share, depth $< 0.1\times$ WETH [%]	2.4	2.4
Stable route share, depth $\geq$ WETH [%]	53.0 (4.6)	61.2 (13.2)
Stable route share, depth $\geq 2\times$ WETH [%]	69.9 (2.3)	70.7 (9.3)
Active pair-days	11,327	17,027
Panel C. Adoption of supported stablecoin after persistent support		
Outcome	First 30 days	First 120 days
Supported stablecoin used, depth $< 0.1\times$ WETH [%]	42.6	50.5
Supported stablecoin used, depth $\geq 0.1\times$ WETH [%]	84.1	89.7
Difference [pp]	$+41.55^{***}$ (4.95)	$+39.17^{***}$ (4.48)
Difference with controls [pp]	$+25.16^{***}$ (5.76)	$+23.47^{***}$ (5.37)
Difference, no stablecoin endpoint [pp]	$+35.76^{***}$ (11.89)	$+40.26^{***}$ (11.66)

Note: The unit is a pair by route scope around the first persistent DAI, USDC, or USDT bridge. Support requires positive prior-calendar deposited capital on both legs in Uniswap v2 or SushiSwap v2 on at least 24 of the next 30 calendar days. Every pair has earlier WETH-mediated routes and no earlier observed stablecoin route. Panel A reports changes from the prior 30 days. Panel B compares the best stablecoin weak-leg capital with WETH weak-leg capital and reports Equation (8). Panel C records use of a stablecoin in the event's support set within 30 or 120 days. Standard errors are two-way clustered by pair and date. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. Deposited capital is an equilibrium pool stock.

Figure 2: Bridge capital accumulates before first use of the supported stablecoin



Note: The sample contains 246 pair-by-route-scope events whose first use of a stablecoin in the event's support set occurs 7 to 119 calendar days after persistent support. Each line is the equal-event mean change from day -7 in $\log(1+\text{deposited capital in USD})$ on the weaker leg. Stablecoin depth is the largest bottleneck among the supported stablecoins; WETH depth is the matched incumbent bottleneck. Capital is lagged to the prior calendar day. Shading gives 95% confidence intervals from standard errors two-way clustered by pair and adoption date.

6 Conclusion

Stablecoins become dominant vehicle currencies on decentralised exchanges through pair formation and local liquidity. Between matched January–June periods in 2024 and 2026, their share among native and stable intermediaries rises from 16.9% to 42.3% by route count and from 32.7% to 76.5% by routed value. Positive and negative substitutions offset inside continuing pairs. Trading grows in continuing pairs that already use stablecoins more intensively, and pairs present in only one period supply the largest additional component. The same rotation appears on routes with one intermediary and on longer routes with several intermediary positions.

The vehicle used at pair entry predicts later use. A 1 pp higher initial stablecoin share predicts +0.74 pp higher stablecoin share over 120 days. Local capital helps explain when another vehicle can compete. First-month stablecoin use is 42.6% when its two-leg bridge has less than one-tenth of WETH depth and 84.1% once it reaches that threshold. Among eventual users, capital on the weaker stablecoin leg rises before the first observed stablecoin route.

Exact pre-transaction prices separate venue competition from vehicle competition. Opening all exact venues finds a better same-vehicle quote on 44.4% of common-support routes. Opening other vehicles and the direct path raises that incidence by 2.0 pp and lowers stablecoin vehicle share by 1.2 pp. Public prices frequently favour another venue while preserving the intermediary asset.

The distinction extends beyond decentralised exchange. Cross-border payment systems still require an institution to convert the source currency into the destination currency. Project Nexus assigns this function to an FX provider, and Project Rialto uses a vehicle currency when direct exchange is unavailable (Bank for International Settlements Innovation Hub, 2024, 2025). Aggregate dominance in such systems reflects both the vehicle selected inside continuing trading relationships and the vehicles inherited by relationships that enter and grow.

A vehicle currency is made at two connected points: when a new trading relationship adopts it and when liquidity on both legs makes the route competitive. Aggregate dominance records both.

A From contract events to economic routes

Route construction begins with the contract and event format of each exchange. We identify pools from the protocol's factory or registry, map contract addresses to tokens, and convert raw integer amounts using the token precision that applied at the time of the trade. Contract addresses define token identity because unrelated tokens can share a symbol and the same economic asset can appear in native and wrapped form. ETH and WETH are consolidated only after this address-level reconstruction; every other token remains distinct unless a stated economic classification combines it. Within each transaction, pool-swap legs are ordered by block, transaction, and log position. The resulting token flows reveal the exchange that occurred. A single connected chain gives one route and therefore one exchange between its pair endpoints; parallel chains identify an order split, and disconnected event groups remain separate. Because one transaction can contain unrelated calls as well as trades across several exchanges, this procedure also prevents two simultaneous routes from being mistaken for one longer route. The output records the leg sequence and pools traversed, together with the pair. It does not infer that sequence from the transaction sender or router label.

Disconnected event groups become more common late in the sample. They comprise 0.8% of reconstructed components in 2024 and 6.2% in 2026. In 2026 the rate is 9.4% among components touching Uniswap v4 and 4.7% among other components. Table 7 uses the internally connected token flow of each group as a separate route. This alternative preserves the principal economic result: the 2026 stablecoin share is 43.7% by count and 77.3% by value, both slightly above the connected-transaction estimates.

Table 7: Disconnected transaction components and vehicle dominance

Sample boundary or estimate	2024	2025	2026
Panel A. Share of reconstructed components excluded			
All components	0.8%	3.7%	6.2%
Components touching Uniswap v4		11.6%	9.4%
Other components		2.7%	4.7%
Panel B. Stablecoin share among native and stable two-leg vehicles			
	2024	2026	Change [pp] (s.e.)
Connected routes, count	16.9%	42.3%	+25.4 (1.05)
Connected routes, value (20% agreement)	32.7%	76.5%	+43.9 (2.02)
Each valid component, count	16.9%	43.7%	+26.8 (1.06)
Each valid component, value (20% agreement)	32.9%	77.3%	+44.4 (1.93)

Note: Panel A reports the share of reconstructed components belonging to transactions with more than one disconnected token-flow group. Uniswap v4 begins during 2025, so its 2024 cell is empty. Panel B compares the principal connected-transaction rule with an alternative that treats each internally well-formed component as a separate route. Stablecoin shares use native and stable two-leg vehicles as the denominator and equal-weight matched January through June dates in 2024 and 2026. Value estimates require source, intermediary, and destination values to agree within 20%. Standard errors use a Newey–West covariance with a 30-day Bartlett bandwidth. The alternative assigns no unobserved handoff between components.

A complete token sequence is sufficient for the route-count sample. Value weighting additionally requires coherent amounts on every leg and a common dollar valuation. A failed valuation removes the observation from the value calculation while preserving the observed route. Appendix F gives the separate exclusion for self-returning routes.

Historical token precision cannot be verified for 13 tokens traded in 22 constant-product pools. Reserve-reconstruction analyses omit those pools. An upper-bound check removes every route involving any of the 13 tokens: the resulting bound is 5,950 routes and \$10.91 million of route value. Only 1,036 of those routes use one of the affected tokens as an intermediary, representing \$1.284 million, or 0.00043% of value on routes whose legs can all be valued. All 13 tokens lie outside the native and stablecoin groups used in the main comparison. The exclusion affects exact reserve reconstruction; the route-composition result is unchanged.

B Pricing models and validation

The counterfactual comparison requires an exact-input quote from each pool's state and the proposed input amount. Because pool families use different invariants and arithmetic, we validate their pricing models separately. For selected executed swaps, we reconstruct the state immediately beforehand and compare predicted with realised output. Curve's amplification coefficient and Balancer's weight ratio are unavailable in the historical subgraph records, so we estimate them on one set of trades and evaluate the resulting quotes on held-out trades. A route enters the cost comparison only when the relevant pool states and pricing models are available before the transaction.

B.1 Constant-product pools

Uniswap v2 and SushiSwap v2 price the product of reserves (Lehar and Parlour, 2025). Let R_{i,p,e^-} and R_{o,p,e^-} denote pool p's input- and output-token reserves immediately before transaction e, Δ_i the token input, f_p the fee rate, and $\varphi_p = 1 - f_p$ the fraction of input retained for pricing. An exact-input swap moves the pool along

$$(R_{i,p,e^-} + \varphi_p \Delta_i)(R_{o,p,e^-} - \Delta_o) = R_{i,p,e^-} R_{o,p,e^-},$$

which inverts in closed form to

$$\Delta_o = \frac{\varphi_p \Delta_i R_{o,p,e^-}}{R_{i,p,e^-} + \varphi_p \Delta_i}, \qquad \varphi_p = \tfrac{997}{1000},$$

the 30 basis point fee both venues charge, applied on chain as the integer ratio above.[7]

A multi-leg route composes these one hop at a time, threading the output token of one pool into the next. The cost comparison evaluates the dollar outputs of the best direct and intermediated

[7] The SushiSwap v2 pool contract implements the same adjustment; see `UniswapV2Pair.sol`.

routes at the same state and dollar input notional,

$$10^4 \Delta C^D_{i,o,k,q,e} = 10^4 \frac{O^D_{i,o,q,e} - O^I_{i,o,k,q,e}}{O^D_{i,o,q,e}}.$$

Both outputs must be computed from one reconstructed pre-transaction state. A realised trade supplies the output of the chosen route alone; the direct and intermediary alternatives require reconstructed states for every pool in the comparison. Quoting each executed constant-product swap against the state immediately before it gives a median absolute error of 0.0000% across 8,024 swaps; 96.7% of quotes are within 1% and 95.2% are within 0.01% of realised output.

B.2 Concentrated liquidity

Uniswap v3 distributes liquidity across price ranges, so executable depth depends on the active range and on the initialized ticks reached by a trade (Adams et al., 2021). Within one active range, let $\widetilde{R}_{i,p,e^-}$ and $\widetilde{R}_{o,p,e^-}$ denote the virtual reserves of the input and output assets. Liquidity L_{p,e^-} and the output-per-input marginal price ρ_{p,e^-} satisfy

$$\widetilde{R}_{i,p,e^-} = \frac{L_{p,e^-}}{\sqrt{\rho_{p,e^-}}}, \qquad \widetilde{R}_{o,p,e^-} = L_{p,e^-}\sqrt{\rho_{p,e^-}}.$$

If f_p is the pool fee and $\varphi_p = 1 - f_p$, an exact input Δ_i returns

$$\Delta_o = \frac{\varphi_p \Delta_i \widetilde{R}_{o,p,e^-}}{\widetilde{R}_{i,p,e^-} + \varphi_p \Delta_i}$$

until the trade reaches the next initialized tick. Range endpoints lie on a geometric grid, and crossing tick j updates active liquidity by its signed change,

$$\sqrt{\rho(j)} = 1.0001^{j/2}, \qquad L_{p,e^-} \mapsto L_{p,e^-} \pm \Delta L_j,$$

with the sign determined by the direction of the trade. The calculation then continues from the new active range until the input is exhausted. The quoter uses the contract's integer Q96 arithmetic, rounding input amounts up and output amounts down.

The active tick map is accumulated from every mint and burn strictly before the validation day, after which same-day liquidity changes and swaps are replayed together by block and log index. Consecutive swaps in one pool supply the price state: the earlier event leaves `sqrtPriceX96` and `tick` immediately before the later swap, while the interleaved event stream supplies active liquidity. We report errors separately by input direction and tick crossing because those cases use different arithmetic branches. Table 8 applies this validation to Uniswap v4. V4 retains concentrated-liquidity pricing for static-fee pools but implements it through a singleton and flash accounting.[8]

[8] See Adams et al., *Uniswap v4 Core*, Sections 1 and 3.

Table 8: Concentrated-liquidity quoter, by direction and tick crossing

Validation group	Swaps	Median	90th pct.	Maximum	Within 1%
token0 in, no tick crossed	2,290	0	0	9.98×10^{-4}	100.0%
token0 in, tick crossed	57	0	0	3.45×10^{-7}	100.0%
token1 in, no tick crossed	1,816	0	1.69×10^{-8}	1.48×10^{-4}	100.0%
token1 in, tick crossed	55	0	5.44×10^{-9}	8.74×10^{-9}	100.0%
Pooled	4,218	0	1.15×10^{-14}	9.98×10^{-4}	100.0%

Note: Absolute quote error in percent of realised output. The sample contains 4,218 realised Uniswap v4 swaps on three dates across 13 pools and nine token pairs, at static fee tiers 0, 7, 8, 100, 500, 10,000 and 20,000. Each quote uses liquidity events ordered strictly before the swap and the price state left by the preceding swap in the same pool.

The validated static fee tiers include 0, 7, 8, and 20,000 hundredths of a basis point. Hook-bearing and dynamic-fee pools are excluded because the raw records do not identify their transaction-level cash flows and realised fees; Appendix E reports their share of v4 trading.

Every direction and tick-crossing row of Table 8 reproduces realised output within 0.001% in the Maximum column.

B.3 StableSwap pools

Curve interpolates between a constant-sum curve, which is flat and charges no slippage, and the constant-product curve of Section B.1.[9] For n tokens with normalized pre-transaction balances R_{j,p,e^-}, amplification A_p, and invariant D_p,

$$A_p n^n \sum_j R_{j,p,e^-} + D_p = A_p D_p n^n + \frac{D_p^{n+1}}{n^n \prod_j R_{j,p,e^-}},$$

where $A_p \to 0$ recovers the constant product and $A_p \to \infty$ recovers the constant sum. The contract solves D_p by Newton iteration and then solves the same invariant for the post-trade output balance $R'_{o,p,e}$ after R_{i,p,e^-} is increased by Δ_i. Defining $D_\Pi = D_p \prod_j D_p/(nR_{j,p,e^-})$, the two loops are

$$D_p \leftarrow \frac{(A_p n \sum_j R_{j,p,e^-} + nD_\Pi)D_p}{(A_p n - 1)D_p + (n+1)D_\Pi}, \qquad R'_{o,p,e} \leftarrow \frac{(R'_{o,p,e})^2 + c}{2R'_{o,p,e} + b - D_p},$$

with b and c formed from the balances of the tokens the swap does not touch. Balances are rescaled to eighteen decimals before either loop runs, because the invariant treats its tokens as interchangeable at par and a six-decimal stablecoin read on an eighteen-decimal scale is off by a factor of 10^{12}. Output is net of the pool fee,

$$\Delta_o = (R_{o,p,e^-} - R'_{o,p,e} - 1)\varphi_p,$$

[9] See Egorov, *StableSwap—Efficient Mechanism for Stablecoin Liquidity*.

Table 9: StableSwap calibration and held-out quote error, by validation day

Day	Pools fitted	Pools excluded	Held-out trades	Median $\widehat{A}_p$	Median error	Within 1%
2021-05-28	9	14	367	12,716	0.030%	100.0%
2022-09-12	11	16	262	20,493	0.029%	100.0%
2023-12-28	20	28	471	23,428	0.039%	98.9%
2025-04-13	42	50	1,207	46,851	0.036%	99.7%

Note: The sample contains four sampled days. $\widehat{A}_p$ is reported in the module's convention, which carries the on-chain precision factor of 100. Median error is the median absolute deviation of the quote from realised output on trades the calibration never saw.

carrying the contract's own off-by-one guard on the raw difference.

A_p is Curve-specific and the standardised subgraph schema drops it, so it is identified from data instead of fetched. Every other quantity in the invariant is observed and every Curve swap reports both amounts, which leaves A_p as the only unknown,

$$\widehat{A}_p = \arg\min_{A_p} Q_{0.90} \left\{ \frac{|Q_p(i \to o, \Delta_i; \chi_{p,e^-}(A_p)) - \Delta_o|}{\Delta_o} \right\},$$

fitted on the first half of a pool-day's trades over a logarithmic grid with a ternary refinement, then evaluated on the trades not used for estimation. A pool-day is retained when the 90th percentile of fitting error is below 1%. Using the upper tail prevents a partial fit from passing: one misspecified pool produced a median fitting error below 0.1% but a 34% median error on the other trades because the invariant fit only part of its activity. Curve has also operated crypto-pools since 2021 that follow the CryptoSwap invariant. Applying StableSwap to these pools yields a best-fit A_p of 3 and a 36% median quote error, so pool-days whose observed trades are inconsistent with StableSwap remain outside this component exercise.

Table 9 reports the held-out fit by validation day.

Across four validation days, held-out median errors range from 0.029% to 0.039%.

B.4 Weighted pools

Balancer weighted pools use a weighted geometric-mean invariant,[10]

$$K_{w,p} = \prod_j R_{j,p,e^-}^{w_{j,p}}, \qquad \sum_j w_{j,p} = 1,$$

which gives the two-token exact-input form

$$\Delta_o = R_{o,p,e^-} \left[1 - \left(\frac{R_{i,p,e^-}}{R_{i,p,e^-} + \varphi_p \Delta_i} \right)^{w_{i,p}/w_{o,p}} \right].$$

[10]See Martinelli and Mushegian, *A Non-Custodial Portfolio Manager, Liquidity Provider, and Price Sensor*.

Only the ratio $w_{i,p}/w_{o,p}$ enters. An equal-weight Balancer pool and a constant-product pool with the same reserves therefore return the same quote. Balancer's weighted-pool contract accepts inputs up to 30% of the input-token reserve; that contract limit governs whether a quote can execute, but it is not the empirical support rule.[11] The counterfactual design applies the common 5% own-price-impact cutoff in Equation (7) to Balancer and every other pool family. The subgraph reports the weight ratio $\theta_p = w_{i,p}/w_{o,p}$ and the swap fee at its head block while balances come from a historical snapshot, so a pool that has repriced itself since is quoted with a current parameter against a past state. Liquidity-bootstrapping pools move weights on a schedule by design and managed pools move them on demand, and a weighted pool's owner can reset the fee at will. What such a pool produces is a small and nearly constant relative bias, which appears directly in the preceding validation: reported parameters reproduce trades in 2024 to a millionth of a basis point and trades in 2022 only to 0.19%. A gap of that size is a fee. The fee enters the quote through $\varphi_p\Delta_i$, so a fee wrong by a basis point moves output by about a basis point, whereas a mistake in the invariant would leave an error that scales with trade size.

We therefore treat both as unknowns to be recovered from the trades a pool actually executed, the move Curve's amplification coefficient needed in Section B.3, and we keep a reported value only where it survives that test. Reconstructing a pool's state from its own event history follows Lehar and Parlour (2025); what the weighted family adds is that two arguments of the pricing function are themselves missing from the historical record, so a reconstructed state alone does not deliver a quote. Every quantity in the weighted invariant other than θ_p and f_p is observed, and every Balancer swap reports both amounts, so with balances reconstructed either parameter is the only unknown. We order a pool-day's trades and assign them alternately to a fitting set $\mathcal{O}_p$ and an evaluation set, and every error reported in Table 10 is measured on the evaluation set. For a trial parameter pair ϑ,

$$\mathcal{E}_p(\vartheta) = Q_{0.90}\left\{ \frac{|Q_p(i \to o, \Delta_i; \chi_{p,e^-}(\vartheta)) - \Delta_o|}{\Delta_o} : e \in \mathcal{O}_p \right\}$$

is the fitting error, defined only when at least nine tenths of $\mathcal{O}_p$ returns a quote at all and undefined otherwise. The upper tail does the gating for the reason it does at Curve, and the coverage condition stops a parameter that prices one corner of a pool's activity from standing for the pool.

Each token pair's weight ratio is fitted on the trades crossing it in either direction, with a trade running from b to a scored at the reciprocal exponent, which is what the invariant implies and which doubles the trades available to pin one number,

$$\widehat{\theta}_{ab,p} = \arg\min_{\theta} \mathcal{E}_p\big(\theta \text{ on } \mathcal{O}_p^{a\to b},\ \theta^{-1} \text{ on } \mathcal{O}_p^{b\to a}\big),$$

searched over a logarithmic grid spanning Balancer's own weight bounds and refined by ternary search in the log exponent, and attempted only for token pairs carrying at least four fitting trades, since one or two are fitted exactly by construction and then read as an exact fit. The fee $\widehat{f}_p$ is

[11] See the input-ratio bound in Balancer's `WeightedMath.sol`.

recovered the same way over the fee range the vault permits. Reading is tried before fitting and the fee before the weights, so each tier adds at most one free scalar and a pool that reproduces its trades on reported parameters is never handed a fitted one,

$$\widehat{\vartheta}_p = \begin{cases} (\theta_p, f_p) & \text{if } \mathcal{E}_p(\theta_p, f_p) \le \bar{e}, \\ (\theta_p, \widehat{f}_p) & \text{else if } \mathcal{E}_p(\theta_p, \widehat{f}_p) \le \bar{e}, \\ (\widehat{\theta}_{\cdot,p}, f_p) & \text{else if } \mathcal{E}_p(\widehat{\theta}_{\cdot,p}, f_p) \le \bar{e}, \\ \text{excluded} & \text{otherwise,} \end{cases} \qquad \bar{e} = 0.1\%. \tag{9}$$

The label a pool carries in the source plays no part in this. A pool-day whose observed trades no tier reproduces is excluded whatever its `poolType` says, and the third tier further requires the fitted ratios to cover nine tenths of the fitting trades, so a pool-day is never accepted on the token pairs that happened to fit and then quoted on the rest.

The frequency with which the parameter search binds measures the sufficiency of reported pool state. Of the 256 pool-days accepted across the twelve validation days, 209 satisfy the error threshold on reported parameters with nothing identified, 43 need the swap fee alone, and 4 need per-token-pair weight ratios. Four fifths of the priced sample therefore carries no fitted parameter, and identification repairs a stale field in a minority of pool-days instead of supplying the invariant.

We set $\bar{e}$ at 0.1% instead of the 1% Curve uses because the achieved errors separate cleanly there. The threshold separates weighted pools, whose held-out errors range from 10^{-9} to 0.004%, from stable, composable-stable, and elliptic-curve pools, whose held-out errors range from 0.05% to 1.08% and have a 99th percentile of 12%. At a 1% threshold, fitted weight ratios would admit the second group. The 0.1% rule excludes it. Curve's failed validation arose from an over-permissive parameter range; here, an over-permissive error threshold would produce the same mistake.

Across the twelve date rows of Table 10, the Backward column returns a median error of 0.0000% and the Pools priced / excluded column identifies 256 of 473 testable pool-days as weighted pools. The other pool-days carry a median 91.8% of tested swap volume and follow different Balancer pricing rules. The exercise therefore validates the model for identified weighted pools, not for Balancer as a whole.

Table 10: Weighted-pool quoter and reserve-timing alternatives

Day	Held-out trades	Backward	Flat	Opening	Pools priced / excluded
2021-04-25	4	4.7×10^{-9}	3.00%	3.94%	1 / 0
2021-09-29	273	2.5×10^{-8}	1.24%	0.80%	18 / 8
2022-03-09	524	1.6×10^{-4}	1.05%	2.66%	25 / 4
2022-08-16	701	1.1×10^{-9}	0.75%	1.04%	33 / 12
2023-01-24	832	2.0×10^{-10}	3.73%	4.71%	44 / 11
2023-07-03	301	7.1×10^{-10}	1.78%	3.32%	26 / 19
2023-12-11	185	2.8×10^{-9}	1.77%	2.91%	17 / 20
2024-05-20	265	3.8×10^{-10}	4.81%	5.17%	15 / 36
2024-10-27	142	1.8×10^{-10}	1.68%	2.24%	12 / 36
2025-04-06	248	2.1×10^{-10}	5.32%	9.48%	19 / 40
2025-09-13	501	2.5×10^{-4}	1.09%	1.71%	27 / 25
2026-02-21	582	8.7×10^{-9}	0.68%	1.90%	19 / 6

Note: Median absolute quote error in percent of realised output, on trades no fit ever saw. *Backward* reconstructs per-swap balances by rolling the day's event sequence back from the closing snapshot, *Flat* keeps the snapshot balances constant across the day, and *Opening* treats the snapshot as the day's opening state and rolls forward. The sample contains twelve days from 2021-04 to 2026-02.

C Reconstructing pool balances before each trade

Every counterfactual quote uses the pool state immediately before the trade, whereas some historical records store balances only at the end of an hour or day. We recover the earlier state by ordering every balance-changing event within the period and unwinding the sequence from the recorded closing balance. The exercises below validate that procedure on selected constant-product and weighted-pool observations. The cost comparison retains only routes whose pools can be reconstructed in this way.

Let $\mathbf{R}_p^{\text{end}}$ be pool p's stored period-end reserve vector and $\mathbf{\Delta}_{p,e}$ the vector of net signed token amounts in event e. The state before event e is the stored value with every later event removed,

$$\mathbf{R}_{p,e}^{\text{pre}} = \mathbf{R}_p^{\text{end}} - \sum_{e' \geq e} \mathbf{\Delta}_{p,e'} = \mathbf{R}_{p,e+1}^{\text{pre}} - \mathbf{\Delta}_{p,e},$$

which is a single reverse pass over the period. The alternative reading takes the previous period's stored value as the current period's opening state and replays forward,

$$\widetilde{\mathbf{R}}_{p,e}^{\text{pre}} = \mathbf{R}_{p,\text{prev}}^{\text{end}} + \sum_{e' < e} \mathbf{\Delta}_{p,e'},$$

which accumulates every unrecorded balance change between the two stored points. Replaying forward returns a median absolute quote error of 11.8% against realised outputs, whereas unwinding backward returns 0.0000%.

Liquidity additions and withdrawals also move constant-product pool balances, so the ordered replay includes them alongside swaps. We unwind the sequence to the start of the period and compare the reconstructed opening balance with an independently observed reserve state,

$$\left\| \left(\mathbf{R}_{p,1}^{\text{pre}} - \mathbf{R}_{p,\text{prev}}^{\text{end}} \right) \oslash \mathbf{R}_{p,\text{prev}}^{\text{end}} \right\|_{\infty} < \tau,$$

Agreement establishes that the event history reproduces the observed reserves for the tested period. The state-dependent sample omits a period when a required event family is incomplete or the balances do not reconcile. Coverage is reported by exchange and year because missing state can be concentrated in actively managed or newly launched pools.

The same logic applies to weighted pools. We treat `poolSnapshots.amounts` as the closing balance and unwind joins, exits, and swaps in their recorded order. Table 10 compares this reconstruction with two alternatives that either hold the closing balance fixed or treat it as the opening state.

The Balancer replay includes joins and exits as well as swaps. Omitting those events assigns intraday liquidity changes to the pricing rule and changes the pool states available for the price comparison.

D Restricting hypothetical trades to observed size–depth combinations

Component-validation samples contain executed swaps and therefore pools deep enough to serve the observed trade. Hypothetical routes can demand much more output relative to pool depth, where the observed errors no longer describe pricing accuracy. We include a hypothetical leg in the cost comparison only when its own price impact is at most 5%, calculated under the pricing rule of the pool family it would traverse.

For constant-product swaps, input relative to reserves provides a transparent size–depth comparison,

$$\iota_{i,p,e^-} = \frac{\Delta_i}{R_{i,p,e^-}},$$

the input amount as a fraction of the input-side reserve. Table 11 reports its distribution in the historical validation sample. The statistic helps calibrate what constitutes a large trade in a constant-product pool, but it does not replace the common own-price-impact rule for concentrated-liquidity, StableSwap, or weighted pools.

In the Pooled row of Table 11, five per cent of input reserves sits between the 90th percentile of 3.29% and the 99th of 14.86%; the same cutoff lies between those columns on seven of the eight date rows. Figure 3 shows the daily distributions. This calculation locates the cutoff in the empirical size distribution. The empirical rule remains a 5% own-price-impact cutoff computed separately under each pool family's pricing function.

Equation (7) applies the same 5% own-price-impact ceiling to every family. The component exercises establish pricing accuracy for realised swaps within their stated samples; they do not

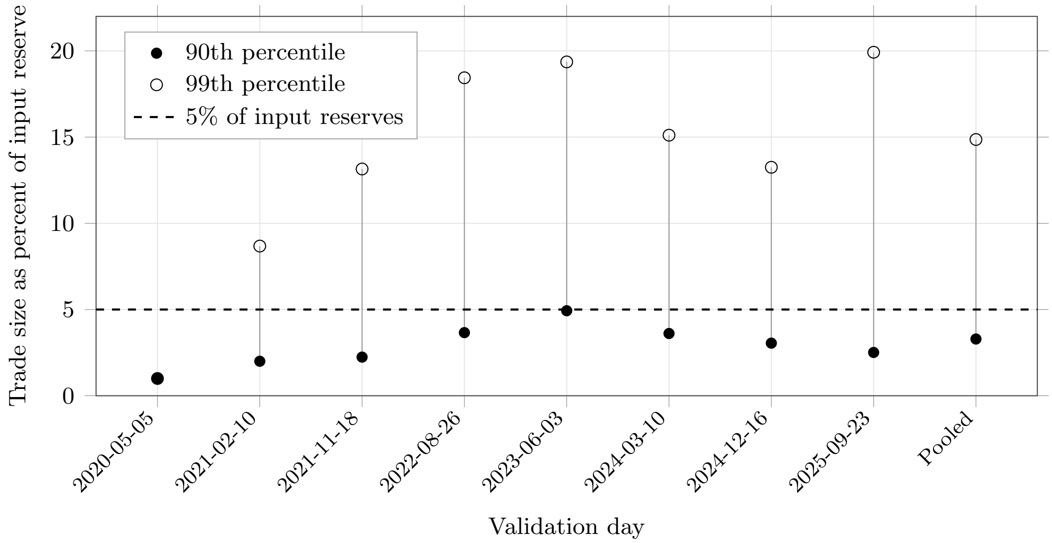
Table 11: Size relative to depth in the validation population

Day	Swaps	Median	90th pct.	99th pct.	99.9th pct.	Maximum
2020-05-05	2	1.00%	1.00%	1.00%	1.00%	1.00%
2021-02-10	141,804	0.19%	2.00%	8.68%	23.17%	99.83%
2021-11-18	107,059	0.10%	2.24%	13.15%	71.64%	100.00%
2022-08-26	79,180	0.48%	3.66%	18.44%	94.13%	100.00%
2023-06-03	159,551	0.85%	4.93%	19.36%	96.59%	100.00%
2024-03-10	215,060	0.41%	3.61%	15.11%	57.81%	100.00%
2024-12-16	132,508	0.31%	3.05%	13.25%	71.93%	100.00%
2025-09-23	97,106	0.24%	2.51%	19.92%	99.11%	100.00%
Pooled	932,270	0.34%	3.29%	14.86%	78.33%	100.00%

Note: Trade size as a percentage of the input-side reserve of the pool traversed, over the constant-product swaps the quoters were validated against. The sample contains eight dates from 2020-05-05 through 2025-09-23.

measure quote error on unexecuted routes.

Figure 3: Constant-product trade size relative to reserves



Note: The statistic is a leg's input amount as a fraction of the input-side reserve in the constant-product pool it traverses. Points mark the 90th and 99th percentiles; the dashed line marks 5% of reserves. The two percentiles coincide on 2020-05-05. The sample contains 932,270 realised swaps on eight dates from 2020-05-05 through 2025-09-23.

E Coverage of the price comparison

Omitting an exchange can change the best direct route, the best intermediated route, or both. Table 12 first describes the volume recorded in nine local source series, including Uniswap v1 as a separate architecture sample and the available Fluid extracts. Volume in the Graph-indexed series is reconstructed from individual swaps using the smaller dollar value of the two legs. This avoids double counting in two series and removes implausible oracle values in thin pools: one Curve pool reports a \$456m day whose swaps sum to \$8, and a constant-product pool with \$255 of reserves reports \$1.36bn. Physical bounds on volume relative to reserves exclude between 0.10% and 0.24% of Graph-indexed pool-days per year. Uniswap v1 is excluded from the route-cost comparison because its forced-routing architecture is studied separately. Fluid supplies volume on only 40 of the 400 sampled dates and no historical TVL field. The table consequently reports source coverage. Complete market shares require a common historical state series for every exchange.

The Uniswap v4 pricing model covers static-fee pools without hooks. Pool-identity reconstruction leaves zero swaps with incomplete or invalid immutable pool attributes across 29,005,863 rows. Pools in this group account for 84.02% of swaps and 99.05% of reported value; the corresponding 2026 shares are 79.89% and 95.57%. Other pools use hooks, dynamic fees, or both, whose transaction-level cash flows are not identified from the raw event. These shares describe the reach of the pricing model. They do not establish the availability of every pre-transaction state.

Curve carries between 9.68% and 18.70% of observed source volume in every year it exists and ranks second in four years, making its coverage economically important. The StableSwap model fits

Table 12: Share of observed decentralised-exchange volume, by source and year

Year	Uni V1	Uni V2	Uni V3	Uni V4	Sushi V2	Sushi V3	Curve	Balancer	Fluid
2020	2.09	75.89	0.00	0.00	10.89	0.00	11.12	0.00	0.00
2021	0.02	31.27	37.97	0.00	17.31	0.00	12.25	1.18	0.00
2022	0.01	7.01	64.07	0.00	4.35	0.00	18.70	5.87	0.00
2023	0.01	9.26	66.81	0.00	1.21	0.03	13.88	8.80	0.00
2024	0.00	11.63	69.06	0.00	0.49	0.01	12.57	5.86	0.38
2025	0.01	4.21	53.10	19.53	0.28	0.01	9.68	1.49	11.70
2026	0.01	2.27	46.02	31.94	0.11	0.16	12.61	0.15	6.73
Pooled	0.05	14.58	54.54	5.30	5.73	0.02	13.37	3.82	2.61

Note: Percentage shares of volume recorded in nine local source series. The weekly calendar contains 400 dates; Fluid is observed on 40 (6 in 2024, 25 in 2025, and 9 in 2026) and supplies no TVL field. Its entries are partial observed-volume contributions, not complete market shares or turnover estimates. The table displays 2020–2026 and pools volume over those years. Uniswap v1 is shown as the separate forced-routing architecture sample and is excluded from the route-cost opportunity set.

621 pool-days and excludes 588 whose 90th-percentile fitting error exceeds 1%. Across 28 sampled dates from 2020-02-11 to 2026-04-28, these exclusions account for 34.2% of the Curve volume assessed, or \$1.83bn of \$5.36bn. Another 2.75% of Curve volume occurs in pool-days with too few trades to estimate and evaluate the amplification coefficient separately. Lowering the minimum from eight trades to four changes annual excluded shares by at most 1.3 pp. Table 13 shows where these exclusions occur.

The exclusions are concentrated in routes involving the native asset. In Panel B of Table 13, the excluded share of native-leg volume exceeds the stable-leg share by at least 33 pp in every sample year and by 54 pp in the 2023 row. Tricrypto pools pairing WBTC and WETH with a stablecoin account for \$818m, or 44.6%, of the \$1.83bn excluded, including seven of the twelve largest excluded pool-days. These pools use CryptoSwap, so the missing coverage is concentrated where Curve supplies depth between ETH and stablecoins.

SushiSwap v3 is omitted from the route-cost comparison because its historical records lack the pool's root price, in-range liquidity, and tick spacing. Its reserves are distributed across ticks, and the balance and weight fields exposed by the common schema do not describe that invariant. The exchange accounts for 0.016% of volume among the seven exchanges used for route costs over the full sample and 0.176% in 2026. Across fourteen sampled days, only 4.1% of its volume occurs on token pairs unavailable on another included exchange, never more than \$37,609 on one day.

Balancer accounts for 3.9% of measured venue volume over the full sample and 8.8% at its 2023 peak. Table 10 validates selected weighted pools. Linear and boosted pools remain outside the quoter. Of 29 sampled composable-stable pools, 23 reduce to two external tokens after removing the pool's own share token, but pricing them also requires historical rate-provider states absent from the source. The priced Balancer sample is therefore confined to weighted pools whose historical state can be reconstructed and whose realised trades are reproduced at one of the tiers of Equation (9).

Table 13: Curve volume outside the StableSwap comparison

Panel A: composition of the exclusion					
Leg served	Pool-days	Volume	Of excluded	Of priced	Of that leg
Native	374	\$1.038bn	56.6%	15.7%	65.2%
Stable	119	\$0.752bn	41.0%	79.9%	21.1%
Other volatile	95	\$0.045bn	2.4%	4.4%	22.2%
Panel B: excluded share of each leg's volume, by year					
Year		Native leg		Stable leg	Other volatile
2021		85.49%		46.91%	15.23%
2022		68.82%		30.21%	50.77%
2023		59.11%		5.51%	56.60%
2024		48.24%		0.94%	14.38%
2025		47.47%		13.96%	10.52%
2026		60.92%		9.04%	20.08%

Note: A pool serves the *native* leg when it contains ETH or a staking derivative, the *stable* leg when every token is stable, including baskets and interest-bearing wrappers, and the *other volatile* leg otherwise. Stablecoin base-pool tokens are classified with the underlying stable assets. The sample contains 28 measured Curve pool-days from 11 February 2020 through 28 April 2026.

F Round-trip exclusion

A route whose first input token equals its last output token,

$$\mathbf{1}\{\, k_0 = k_{m(r)} \,\},$$

does not represent a conversion between distinct endpoint assets. We exclude these routes from the dominance measures. The excluded population can contain cyclic arbitrage and other self-returning activity; the endpoint rule does not classify every route as arbitrage or wash trading. Multi-trade arbitrage can be bundled atomically (Daian et al., 2020), but identifying trader intent would require additional evidence.

Table 14 reports the distribution across sampled days; Figure 4 shows the day-level observations.

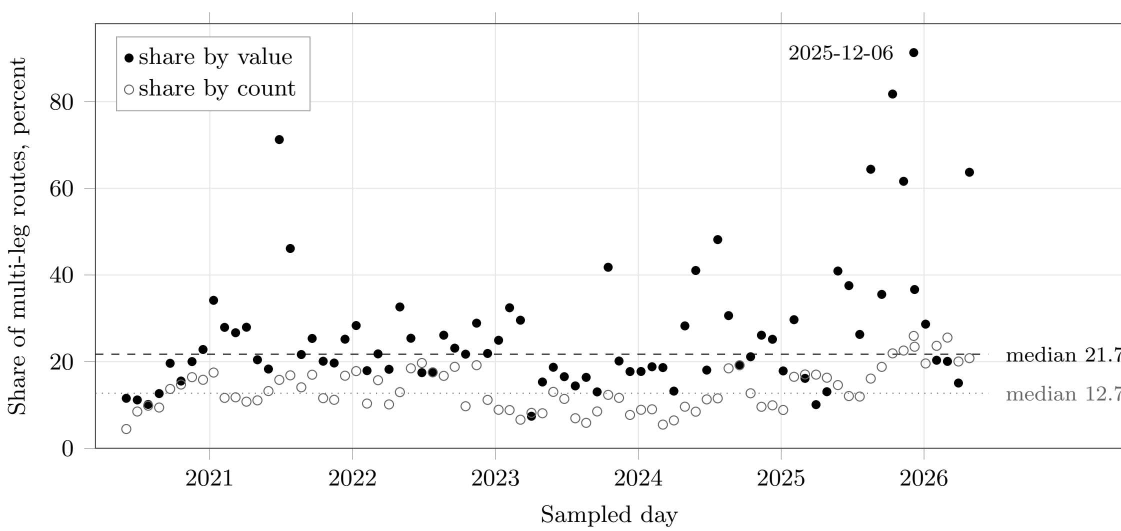
Across the quantile rows of Table 14, the Share by value column exceeds Share by count by a factor between 1.7 and 3.5, consistent with a population in which large self-returning routes carry disproportionate value. Round trips appear on every sampled day; their minimum count share is 4.45%. Round trips are excluded before the count- and value-weighted dominance measures are formed. Figure 4 plots their count and value shares among multi-leg routes and identifies the date of the maximum.

Table 14: Round-trip share of multi-leg routing, across 79 sampled days

Quantile across days	Share by count	Share by value
Minimum	4.45%	7.40%
25th percentile	9.60%	17.73%
Median	12.70%	21.72%
75th percentile	17.04%	29.56%
90th percentile	20.03%	46.11%
Maximum	25.93%	91.31%
2020 median	11.77%	14.08%
2021 median	13.21%	25.35%
2022 median	16.72%	21.89%
2023 median	8.51%	17.71%
2024 median	9.60%	21.13%
2025 median	16.75%	36.08%
2026 median	20.80%	20.32%

Note: Round trips as a share of multi-leg routes. The sample contains 79 dates from 1 June 2020 through 27 April 2026 and 13,160,047 routes, of which 2,488,399 are multi-leg and 351,553 are round trips. Dates with fewer than 200 multi-leg routes are omitted.

Figure 4: Round-trip contamination of multi-leg routing, day by day



Note: Each mark is one sampled date. *Share by value* weights round-trip routes by their notional; *share by count* gives each route equal weight. Both shares use multi-leg routes as the denominator. Dashed and dotted lines mark the respective medians, and the highest point is labelled. Dates with fewer than 200 multi-leg routes are omitted. The sample contains 79 dates from 1 June 2020 through 27 April 2026 and 2,488,399 multi-leg routes.

G Additional composition and network results

The full route panel supports two complementary measures. Intermediary-position share allocates all observed intermediary work across currencies. Route-participation share records the fraction of routes containing a currency and can sum above one across currencies because a longer route may contain several intermediaries. Panel A of Table 15 reports both measures for the native and stablecoin families. Panel B compares realised use with each named currency's position in the trading graph.

Table 15, panel A, shows the same stablecoin rotation under both all-route normalisations. Panel B shows a different movement in graph position. WETH remains the first-ranked betweenness node in every year, even as USDC and USDT gain realised intermediary use. The result separates the existence of leg connections from the depth and execution conditions that determine route allocation.

For a pair active on date t with no stablecoin vehicle, define $H_{p,t+30}$ as an indicator that a stablecoin route appears within 30 days. The rolling-hazard specification is

$$H_{p,t+30} = \alpha + \beta_N \log(1 + N_{p,t}) + \beta_A \log(1 + A_{p,t}) + \mu_d + \varepsilon_{p,t+30}, \tag{10}$$

where $N_{p,t}$ is origin-date route count, $A_{p,t}$ is pair age, and μ_d absorbs month-day effects. The mixed-vehicle analysis compares vehicles inside the same set defined by pair, date, and route class and relates each vehicle's route share to its network reach outside that set.

Table 16 shows that one log point of origin-date route count predicts +3.5 pp higher probability of stablecoin arrival within 30 days, and pair age predicts another +1.6 pp. Once native and stable vehicles coexist, same-day and prior-30-day network reach predict route allocation.

USDC and USDT together account for 92.1% of the increase in stablecoin route-count share. The direct endpoint decomposition identifies USDT inside the high-value stable-to-stable channel. Table 17 reports the supplementary role-specialisation ratio, defined as intermediary share divided by endpoint share.

On matched dates, USDT's intermediary share minus its endpoint share rises from −7.13 pp to +8.14 pp. Its intermediation-intensity ratio rises from 1.06 to 1.23 by count and from 0.59 to 1.42 by value. The observed route can begin after the user's broader instruction when the executor supplies inventory, so the direct additive decomposition remains the issuer-level result.

Let $I_{a,t}$ be currency a's share of the day's intermediary uses and $D_{a,t}$ its share of route-endpoint appearances, both in pp. We estimate

$$I_{a,t} = \alpha + \beta D_{a,t} + \mathbf{A}'_a\theta + \mu_t + \eta_a + \varepsilon_{a,t}, \tag{11}$$

where $\mathbf{A}_a$ contains asset-class indicators, μ_t date effects, and η_a currency effects. Currency effects absorb the time-invariant class indicators, so the two do not appear together in the final specification.

Table 15: All-route participation and trading-network position

Panel A. All complete route lengths

Year	Currency family	Intermediary-position share [%]	Route-participation share [%]	Routes using family
2020	Native	70.5	77.1	2,777,213
	Stablecoin	26.1	26.9	968,919
2024	Native	75.3	79.5	6,894,736
	Stablecoin	17.2	17.6	1,522,324
2026 H1	Native	42.0	50.1	2,023,656
	Stablecoin	41.9	46.1	1,865,152

Panel B. Named currencies in the trading graph

Currency	Year	Betweenness (rank)	Intermediary-position share [%]	Route-participation share [%]
WETH	2020	0.961 (1)	69.2	75.9
	2024	0.985 (1)	76.2	80.5
	2026 H1	0.927 (1)	42.3	50.5
USDC	2020	0.032 (3)	11.2	12.3
	2024	0.039 (3)	8.1	8.6
	2026 H1	0.104 (2)	21.1	25.2
USDT	2020	0.074 (2)	11.1	12.2
	2024	0.055 (2)	6.8	7.2
	2026 H1	0.104 (3)	16.3	19.5

Note: Panel A uses all coherent complete routes that do not return to their starting asset. Intermediary-position shares use all intermediary positions as the denominator. Route-participation shares use complete routes as the denominator and may sum above 100% across currency families. The 2020 and 2024 rows use full calendar years; 2026 H1 covers January through June. Panel B builds one undirected trading graph for each year from legs observed on the fifteenth day of each month. An edge records at least one leg in either direction during the sampled year. Betweenness is unweighted and approximated from 256 source nodes with a fixed seed. Realised-use shares in Panel B use the same sampled dates and retain all route lengths. Ranks are shown in parentheses.

Table 16: Stablecoin arrival and vehicle network reach

Model	Outcome	Regressor	Coefficient / (s.e.) [pp]	Obs. / clusters
Turn-on	Stable appears	Baseline route count	-2.7^{***} (0.4)	94,260 / 181
Turn-on	Stable appears	Direct-route share	$+27.1^{***}$ (6.4)	94,260 / 181
Turn-on	Stable appears	Complex-route share	$+20.3^{***}$ (5.1)	94,260 / 181
Thin/direct	Stable appears	Direct share × thinness	$+3.1^{***}$ (0.9)	94,260 / 181
Leader switch	Stable becomes leader	Baseline route count	-1.1^{***} (0.2)	94,260 / 181
Rolling hazard	Stable appears within 30d	Origin route count	$+3.5^{***}$ (0.7)	1,690,478 / 303
Rolling hazard	Stable appears within 30d	Pair age	$+1.6^{***}$ (0.2)	1,690,478 / 303
Mixed risk set	Vehicle route share	Same-day reach	$+9.4^{***}$ (2.9)	27,852 / 363
Mixed risk set	Vehicle route share	Prior-30d reach	$+7.8^{***}$ (2.5)	27,852 / 363
Issuer split	Vehicle route share	USDC × 2026	$+53.0^{**}$ (22.5)	27,852 / 363
Issuer split	Vehicle route share	USDT × 2026	$+88.2^{***}$ (24.2)	27,852 / 363

Note: Rolling-hazard rows estimate Equation (10) on active native-only pair-days with complete 30-day support. Turn-on and leader-switch rows compare matched January–June observations defined by pair, date, and route class in 2024 and 2026. Mixed-set rows compare native and stable vehicles within the same observed set defined by pair, date, and route class. Coefficients appear above clustered standard errors in parentheses and are reported in pp. Reach, route count, and pair age enter as $\log(1+x)$. The designs use month-day or risk-set fixed effects as appropriate and cluster by pair and date. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. The estimates are predictive associations.

Table 17: USDT role specialisation relative to observed endpoint demand

	2024	2026
Count excess-use ratio (2024 full year; 2026 January–June)	1.06	1.23
Value-weighted excess-use ratio (2024 full year; 2026 January–June)	0.59	1.42
Paired January–June intermediary minus route-endpoint share [pp]	−7.13	+8.14

Note: Intermediary and endpoint shares are formed across native, staked-native, stable, and imported currencies on complete routes that do not return to their starting asset. Direct routes enter the endpoint denominator. Intermediation intensity is intermediary share divided by endpoint share; the gap is their difference. Ratios use full-year 2024 and January–June 2026, while the share gap uses matched January–June dates. Dollar-weighted measures apply the 20% value-agreement rule. The ratio measures specialisation, not aggregate importance.

Table 18: Intermediary role within a day

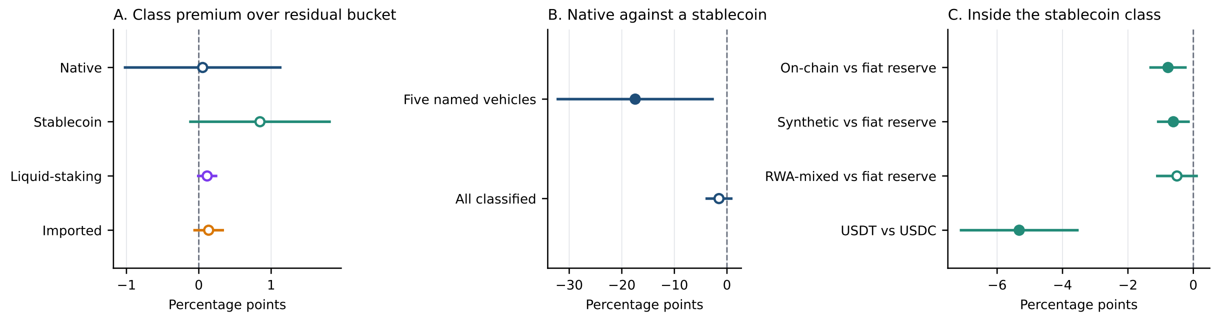
	Intermediary episode share ($I_{a,t}$) [pp]			
	(1) Pooled	(2) Date FE	(3) + demand	(4) + currency FE
Native currency	+34.56	+34.55	+0.05	—
	(23.04)	(23.04)	(0.56)	—
Stablecoin	+2.42*	+2.42*	+0.85*	—
	(1.32)	(1.32)	(0.50)	—
Own endpoint-demand share ($D_{a,t}$)	—	—	+1.59***	+0.98***
	—	—	(0.03)	(0.30)
R^2	0.485	0.485	0.953	0.208
Observations	1,828,862	1,828,862	1,828,862	1,828,862
Currencies	304,572	304,572	304,572	304,572
Dates	2,259	2,259	2,259	2,259
Date fixed effects	No	Yes	Yes	Yes
Currency fixed effects	No	No	No	Yes

Note: The table estimates Equation (11) on 1,828,862 currency-days over 304,572 currencies and 2,259 dates. The outcome and demand regressor are measured in pp. Coefficients appear above standard errors in parentheses clustered by date and currency. The final column absorbs currency and date effects jointly, leaving class indicators unidentified. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 18 reports a demand coefficient of 1.587 across currencies within a date and 0.985 within a currency after absorbing date and currency effects. Figure 5 reports the class and issuer contrasts.

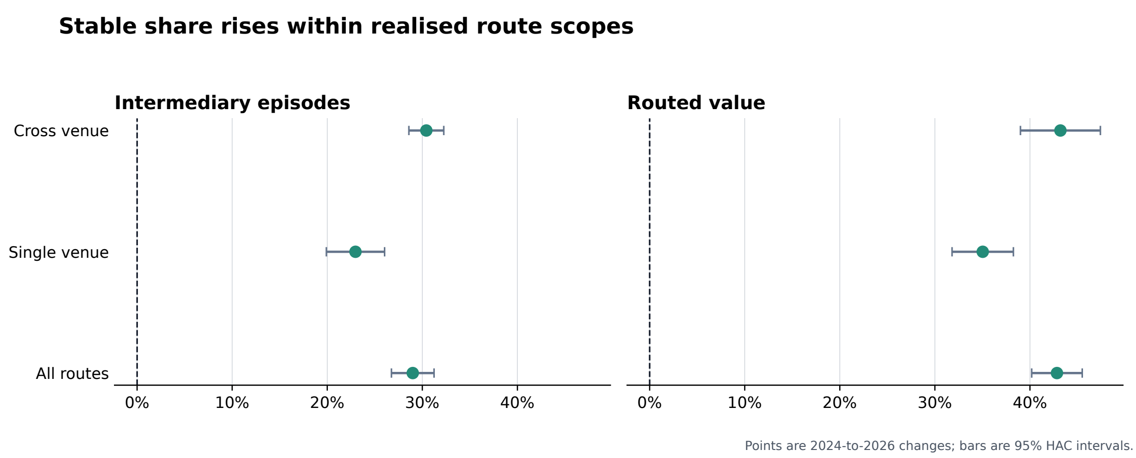
Figure 6 shows that cross-exchange routes grow from 2.0% of route count in 2020 to 57.2% in 2026 and account for 79.1% of routed value in 2026. Within a date, cross-exchange routes carry +14.18 pp more stablecoin share by routed value than single-exchange routes; the count difference is +3.73 pp. Exchange span is selected by the router, so this comparison is descriptive.

Figure 5: Within-day intermediary role, conditional on endpoint demand



Note: Each row is one coefficient from Equation (11) with date fixed effects. Bars are 95% confidence intervals. Panel A compares asset classes with the residual unclassified group, Panel B compares the native asset with stablecoins, and Panel C compares stablecoin backing regimes and USDT with USDC. The panels use separate horizontal scales.

Figure 6: Stablecoin intermediary share rises within single- and cross-exchange routes



Note: The sample contains complete routes that do not return to their starting asset, including routes with more than two legs. Each route contributes once for every intermediary it uses. Stablecoin shares use all intermediary types as the denominator. Count and dollar-weighted estimates compare matched January–June dates in 2024 and 2026. Dollar-weighted measures apply the 20% value-agreement rule.

H Secondary route and liquidity comparisons

This section collects comparisons that complement the central pair decomposition and bridge-depth evidence. They describe changes around router releases, differences across venue pricing families, the composition of capital growth before first use, subsequent comovement between route use and capital, and provider activity in Uniswap v3 and v4. The event dates and pool states are equilibrium outcomes, so these estimates describe patterns in the data.

H.1 Router releases and venue pricing families

Table 19 compares route structure before and after three public Uniswap router releases. Cross-exchange routing expands around the first two releases, while the incidence of intermediation and route length move little. Table 20 divides routes by the pricing family used on every leg. The stablecoin rotation is present among routes priced entirely by the constant-product family, which carries 84.8% of vehicle-currency intermediary positions in 2026.

H.2 Capital composition and later adjustment

Table 21 separates newly active pools from capital growth in pools already supporting the route. Newly active pools hold 7.5% of stablecoin weak-leg capital on the first-use day; 92.5% lies in pools active one week earlier. Table 22 follows the same local bridge across exact 30- and 120-day horizons. Current route share predicts later depth growth, and current depth predicts later route-share growth. Both directions remain descriptive because expected use and capital can adjust together.

Table 23 extends the description to provider activity. The broad collection of outcomes is useful for locating adjustment, while the shared demand and capital shocks in these data keep it outside the paper's central evidence.

H.3 Uniswap v4 settlement and provider activity

Uniswap v4 records a complete pool route inside one singleton and settles the final net balances at the end of the call. Table 24 relates its internal routing and net settlement to later provider activity, and Table 25 shows how those associations vary with persistent volatility. The remaining comparisons through Table 27 use within-day and cross-protocol variation while preserving an observational interpretation.

Table 19: Route structure around public router releases

	Indirect	Intermediated	Cross-exchange	Legs	Exchanges	Over two legs
Panel A: Auto Router, September 16, 2021 (60 days each side)						
Before	22.7	20.8	8.5	2.08	1.10	6.6
After	18.9	15.1	11.0	2.10	1.11	8.1
Change	−3.7	−5.7	+2.5	+0.01	+0.01	+1.5
Panel B: Cross-version Auto Router, December 16, 2021 (60 days each side)						
Before	18.5	13.9	11.3	2.11	1.11	8.4
After	18.6	14.8	14.4	2.16	1.16	11.3
Change	+0.2	+0.8	+3.1	+0.05	+0.05	+2.9
Panel C: Universal Router, November 17, 2022 (60 days each side)						
Before	19.8	13.2	26.4	2.19	1.22	13.1
After	19.5	12.5	24.4	2.20	1.20	12.0
Change	−0.3	−0.7	−2.0	+0.01	−0.03	−1.1

Note: Symmetric windows cover 60 calendar days on either side of each release and exclude the release date. Indirect is the share of economic routes with more than one leg; Intermediated is the share using a third asset; Cross-exchange is the share of intermediated routes using more than one exchange. Shares and changes are in pp. The releases were available to every pair at the same time, so the table reports route composition around the dates.

Table 20: Vehicle role specialisation by venue pricing family

	All venues		Constant product		Balancer		Curve	
Year	Count	Value	Count	Value	Count	Value	Count	Value
Panel A: Stablecoins								
2020	1.45	0.42	1.50	0.42	no routes		no intermediation	
2021	1.18	0.89	1.20	0.92	2.09	1.43	no intermediation	
2022	1.46	1.08	1.50	1.12	1.91	1.00	no intermediation	
2023	1.89	0.94	1.95	1.00	1.46	0.37	no intermediation	
2024	1.27	0.84	1.28	0.78	1.01	0.27	no intermediation	
2025	1.23	1.21	1.23	1.20	0.70	0.09	no intermediation	
2026	1.41	1.24	1.42	1.33	1.75	1.68	no intermediation	
Panel B: Native asset								
2020	0.91	1.56	0.90	1.37	no routes		no intermediation	
2021	0.94	1.13	0.94	1.09	0.24	0.66	no intermediation	
2022	0.85	0.96	0.85	0.93	0.35	0.90	no intermediation	
2023	0.87	1.16	0.88	1.08	0.50	0.53	no intermediation	
2024	0.95	1.20	0.95	1.29	0.22	0.20	no intermediation	
2025	0.89	0.67	0.92	0.85	0.19	0.19	no intermediation	
2026	0.77	0.58	0.80	0.62	0.51	1.30	no intermediation	

Note: Role specialisation is an asset type's share of intermediary activity divided by its share of route-endpoint demand, formed within each venue family. A route enters a family only when every leg uses a venue in that family. The constant-product family contains Uniswap v2 through v4 and SushiSwap v2 and v3. Uniswap v1 enters the separate forced-routing sample. Balancer is listed separately because its pools use both weighted and stable invariants. Values use all routes with positive dollar support. The ratio measures vehicle use relative to endpoint use; it is distinct from aggregate vehicle share.

Table 21: Pool activation and capital growth before stablecoin route use

Outcome	Stablecoin	WETH	Difference
Weak-leg deposited capital [log change]	$+0.76^{***}$ (0.16)	$+0.06$ (0.07)	$+0.70^{***}$ (0.17)
Active pool count [log change]	$+0.04^{***}$ (0.01)	-0.01^{*} (0.01)	$+0.05^{***}$ (0.01)
Any newly active pool [pp]	$+7.32^{***}$ (1.66)	$+0.41$ (0.41)	$+6.91^{***}$ (1.72)
New-pool share of adoption capital [pp]	$+7.47^{***}$ (1.70)	$+0.41$ (0.41)	$+7.14^{***}$ (1.77)
Continuing-pool capital [log change]	$+0.23^{***}$ (0.07)	$+0.07^{*}$ (0.04)	$+0.16^{**}$ (0.08)

Note: The sample contains the bridge events in Figure 2. Stablecoin and WETH outcomes use each vehicle's weaker leg. A newly active pool has positive prior-calendar deposited capital on day zero and none on day -7; a continuing pool has positive capital on both dates. Each cell is an equal-event mean. Parentheses contain standard errors two-way clustered by pair and adoption date. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 22: Route use and subsequent local bridge depth

Feedback equation	30 days	120 days
$R_{b,t} \to \Delta B_{b,t+h}$, pooled [log points]	$+0.020^{***}$ (0.005)	$+0.097^{***}$ (0.013)
$R_{b,t} \to \Delta B_{b,t+h}$, stable [log points]	$+0.024^{***}$ (0.005)	$+0.115^{***}$ (0.012)
$B_{b,t} \to \Delta R_{b,t+h}$, pooled [pp]	$+1.262^{***}$ (0.189)	$+1.595^{***}$ (0.271)
$B_{b,t} \to \Delta R_{b,t+h}$, stable [pp]	$+1.773^{***}$ (0.228)	$+2.463^{***}$ (0.292)
Rows / pairs / dates	51,240 / 503 / 303	18,304 / 340 / 123
Asset and date effects		Yes
Two-way clustered covariance		Pair and date

Note: For source–vehicle–destination bridge b, route use is the vehicle share of five-vehicle route activity and bridge depth is $\log(1 + \min\{K_{iv,t}, K_{vj,t}\})$. Changes compare exact future dates with the origin date. Route-use rows report later depth growth for 10 pp more current route share. Bridge-depth rows report later route-share growth for one log point more current depth. Regressions absorb vehicle and origin-date effects, control for the other current state, weight by route count, and cluster by pair and date. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 23: Provider activity following stablecoin route–capital gaps

Margin	Outcome ($Y_{c,t+h}$)	Horizon	Effect of +10 pp in $G^S_{c,t}$	Obs.	Clusters
V2 stock	Capital share change [pp]	120d	$+1.4^{***}$ (0.2)	10,595	2,119
V2 stock	Log deposited capital [log points]	120d	$+0.205^{***}$ (0.031)	10,595	2,119
Portfolio rank	Capital-rank catch-up [rank positions]	120d	$+0.225^{***}$ (0.036)	10,595	2,119
Same pool	Same-pool capital [log points]	120d	+0.007 (0.009)	174,712	2,113
Pool footprint	Incumbent-pool capital [log points]	120d	$+0.124^{**}$ (0.054)	10,417	2,117
Rent incidence	V3 fee yield [log points]	120d	−0.003 (0.005)	208,360	189
V3 churn	Mint events [log points]	30d	$+0.679^{***}$ (0.037)	11,195	2,239
V3 churn	Burn events [log points]	30d	$+0.642^{***}$ (0.035)	11,195	2,239
V3 churn	Net mint-burn balance [events]	30d	$+0.018^{***}$ (0.002)	11,195	2,239
V3 activity-controlled	Provider-day activity [log points]	120d	$+0.060^{***}$ (0.010)	11,195	2,239
V4 accounting	Multi-leg transaction share [pp]	same day	$+5.4^{***}$ (0.6)	2,599	522
V4 accounting	Internal same-asset share [pp]	same day	$+4.5^{***}$ (0.4)	2,599	522
V4 accounting	Gross-to-net reduction share [pp]	same day	$+2.3^{***}$ (0.4)	2,599	522
V4 LP flow	Gross vehicle-side flow [log points]	120d	$+0.309^{***}$ (0.041)	2,592	521
V4 LP flow	Add flow [log points]	120d	$+0.240^{***}$ (0.036)	2,592	521
V4 LP flow	Remove flow [log points]	120d	$+0.401^{***}$ (0.047)	2,592	521
V4 activity-controlled	LP actions [log points]	120d	$+0.300^{***}$ (0.024)	2,599	522
V4 activity-controlled	Sender-days [log points]	120d	$+0.126^{***}$ (0.016)	2,599	522

Note: Each cell reports the association with a 10 pp larger stablecoin route-share minus deposited-capital-share gap, with clustered standard errors in parentheses. Capital-share changes contain mechanical mean reversion because origin capital share enters both the gap and the future change. Log outcomes use $\log(1+q)$. Vehicle-day specifications absorb vehicle and origin-date effects; pool specifications absorb pool–vehicle and origin-date effects. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 24: Uniswap v4 internal settlement and subsequent liquidity provision

Flash-accounting proxy ($M_{c,t}$)	Future LP flow [log points]	Vehicle-linked TVL [log points]	LP actions [log points]	Flow narrow/medium [pp]	Flow broad [pp]	Action narrow/medium [pp]	Action wide/very-wide [pp]
Internal same-asset share	$+0.035^{*}$ (0.020)	$+0.265^{***}$ (0.031)	$+0.071^{***}$ (0.013)	$+1.398^{***}$ (0.137)	-1.398^{***} (0.137)	-2.460^{***} (0.384)	$+2.889^{***}$ (0.354)
Multi-leg transaction share	$+0.076^{**}$ (0.032)	$+0.123^{***}$ (0.018)	$+0.058^{***}$ (0.017)	$+1.165^{***}$ (0.104)	-1.165^{***} (0.104)	-1.867^{***} (0.203)	$+1.808^{***}$ (0.224)
Gross-to-net reduction share	$+0.139^{***}$ (0.041)	+0.026 (0.031)	$+0.044^{**}$ (0.019)	$+1.223^{***}$ (0.200)	-1.223^{***} (0.200)	+0.362 (0.373)	+0.021 (0.328)
Observations / date clusters							2,015 / 403
Asset and date effects							Yes
Origin-day activity controls							Yes

Note: Each cell reports the association between a 10 pp increase in the row's v4 accounting measure and a future provider outcome, with standard errors in parentheses. The unit is a vehicle-day. Specifications absorb vehicle and origin-date effects, include origin-day activity, flow, reported-liquidity, action, and range-composition controls, and cluster by origin date. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 25: Uniswap v4 participation under persistent volatility

	Incumbent-origin actions, days 1 to 30	First-active origins, days 31 to 120
Internal same-asset share	$+0.067^{***}$	$+0.160^{***}$
	(0.025)	(0.030)
$\times$ lagged 30-day WETH volatility [1 SD]	-0.078^{***}	$+0.318^{***}$
	(0.024)	(0.039)
Observations / date clusters		1,107 / 223
Vehicle and date effects		Yes
Origin-day activity controls		Yes
Vehicle and control volatility slopes		Yes

Note: Each cell reports the internal-routing association at the named volatility state, with standard errors in parentheses. Outcomes are future log action counts by previously active or first-active transaction origins. Specifications absorb vehicle and origin-date effects, allow vehicle-specific volatility slopes, interact volatility with the origin-day controls, and cluster by origin date. Transaction origins proxy for account participation. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 26: Stablecoin route–capital gaps, v4 settlement, and LP repositioning

Interaction term ($G^S_{c,t} \times M_{c,t}$)	Vehicle-linked TVL [log points]	LP actions [log points]	Narrow/medium ranges [pp]	Wide/very-wide ranges [pp]
Stable gap $\times$ internal same-asset share	$+0.292^{***}$	$+0.082^{***}$	-6.531^{***}	$+6.189^{***}$
	(0.079)	(0.027)	(0.720)	(0.657)
Stable gap $\times$ multi-leg transaction share	$+0.225^{***}$	$+0.132^{***}$	-4.539^{***}	$+4.438^{***}$
	(0.035)	(0.013)	(0.442)	(0.381)
Stable gap $\times$ gross-to-net reduction share	-0.667^{***}	-0.121^{***}	$+4.411^{***}$	-4.405^{***}
	(0.105)	(0.043)	(1.261)	(1.126)
Observations / date clusters				1,209 / 403
Asset and date effects				Yes
Origin-day activity controls				Yes

Note: Each cell reports the interaction of a 10 pp larger stablecoin route–capital gap with a 10 pp larger v4 accounting measure. The unit is a stablecoin vehicle-day. Specifications use the 120-day horizon, absorb vehicle and origin-date effects, include origin-day activity and provider controls, and cluster by origin date. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 27: Stablecoin route–capital gaps and liquidity provision in v4 relative to v3

Provider outcome	Horizon	V4-minus-V3 effect	Obs. / dates
LP actions [log points]	120 days	$+0.320^{***}$ (0.013)	5,198 / 522
Active provider origins [log points]	120 days	$+0.115^{***}$ (0.010)	5,198 / 522
Gross vehicle-side flow [log points]	120 days	+0.029 (0.022)	5,188 / 521
Add-side flow [log points]	120 days	+0.012 (0.023)	5,188 / 521
Remove-side flow [log points]	120 days	$+0.046^{**}$ (0.022)	5,188 / 521
Narrow/medium flow share [pp]	30 days	$+1.126^{***}$ (0.133)	5,188 / 521
Reported liquidity [log points]	120 days	$+0.637^{***}$ (0.042)	1,856 / 188
Pool footprint [log points]	120 days	$+0.275^{***}$ (0.022)	1,856 / 188
Vehicle-date and protocol effects			Yes
Origin provider controls			Yes

Note: Each row reports the v4-minus-v3 difference associated with a 10 pp larger stablecoin route–capital gap, with standard errors in parentheses. The stacked unit is a protocol–vehicle–date row on dates with observed v4 singleton activity. Specifications absorb vehicle-date and protocol effects, include the corresponding protocol's origin-day provider controls, and cluster by date. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

I Additional liquidity-provider estimates

Table 28 separates actions by previously active and first-active transaction origins. Table 29 compares routing-linked participation in v3 and v4 on the same vehicle-dates. Table 30 reports vehicle-token flow outcomes that accompany Table 26. Tables 31, 32, and 33 report every horizon and outcome behind Table 27.

Table 28: V4 internal settlement and transaction-origin participation

Flash-accounting proxy ($M_{c,t}$)	Newly active origins, days 1 to 30	Incumbent-origin actions, days 1 to 30	First-active origins, days 31 to 120	Incumbent-origin actions, days 31 to 120
Internal same-asset share	-0.065 (0.028)	$+0.086^{***}$ (0.024)	$+0.153^{***}$ (0.032)	$+0.076^{***}$ (0.018)
Multi-leg transaction share	$+0.061^{*}$ (0.023)	$+0.045^{*}$ (0.017)	$+0.034$ (0.019)	-0.006 (0.014)
Gross-to-net reduction share	-0.221^{***} (0.055)	-0.053 (0.034)	$+0.004$ (0.031)	-0.115^{**} (0.036)
Observations / date clusters				1,107 / 223
Vehicle and date effects				Yes
Origin-day activity controls				Yes

Note: Each cell reports the response to a 10 pp increase in the row's V4 accounting proxy, with standard errors in parentheses. The unit is a vehicle-day. The primary sample excludes modify-liquidity updates with zero liquidity change. An incumbent origin was active for the same vehicle during the preceding 180 days. A newly active origin in days 1 to 30 was absent from that prior window. A first-active origin in days 31 to 120 was absent both from the prior window and from days 1 to 30. All specifications absorb vehicle and origin-date effects, include origin-day swap activity, vehicle-token-side LP flow, vehicle-linked reported liquidity, LP actions, and range composition, and cluster by origin date. P-values are Holm-adjusted across the twelve displayed estimates. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. Transaction origins proxy for account participation, not beneficial ownership of LP positions. The estimates are predictive associations.

Table 29: Internal routing and subsequent participation in V3 and V4

	Incumbent actions, days 1 to 30	First-active origins, days 31 to 120
Panel A. Mature common calendar		
V3 internal-routing slope	−0.003	−0.051
	(0.048)	(0.033)
V4 internal-routing slope	+0.075	−0.045
	(0.029)	(0.027)
V4 minus V3	+0.078	+0.006
	(0.052)	(0.043)
Panel B. Interaction with persistent volatility		
V3 routing × volatility	−0.072	+0.082
	(0.051)	(0.045)
V4 routing × volatility	−0.042	+0.176
	(0.026)	(0.033)
V4 minus V3	+0.031	+0.095
	(0.058)	(0.058)
Panel C. V4 minus V3 with 90-day history		
Mature dates	+0.070	−0.015
	(0.054)	(0.040)
Early dates	+0.115	+0.043
	(0.098)	(0.023)
Pooled dates	+0.188***	+0.274***
	(0.059)	(0.037)
Primary vehicle-days / dates		1,107 / 223
Vehicle and date effects		Yes
Protocol-specific activity controls		Yes

Note: Each coefficient is in log points for 10 pp more internal same-asset routing. Standard errors are in parentheses. Panel A reports protocol-specific slopes and their difference from 23 July 2025 through 2 March 2026, requiring 180 prior days to classify incumbent origins. Panel B interacts routing with one standard deviation more lagged 30-day mean realised WETH volatility on that calendar. Panel C uses a 90-day prior window and separates the mature calendar from 24 April through 27 July 2025; the Pooled dates row combines both periods. The unit is a vehicle-day with positive vehicle transaction counts in both protocols. Specifications first difference V4 and V3 outcomes for the same vehicle-day, absorb vehicle and origin-date effects, include protocol-specific routing-activity and prior-action controls, and cluster by origin date. State specifications also include vehicle-specific and protocol-control-specific volatility slopes. Stars appear only on V4-minus-V3 rows and use Holm adjustment across the two outcomes within each sample. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. V3 participation uses nonzero mint and burn events; V4 participation uses nonzero modify-liquidity events. Transaction origins proxy for account participation, not beneficial ownership of LP positions. The estimates are predictive protocol comparisons.

Table 30: Stablecoin shortfalls, V4 flash accounting, and LP-flow sides

Interaction term ($G^S_{c,t} \times M_{c,t}$)	Gross LP flow [log points]	Add-side LP flow [log points]	Remove-side LP flow [log points]
Stable gap × internal same-asset share	−0.327***	−0.303***	−0.354***
	(0.057)	(0.057)	(0.061)
Stable gap × multi-leg transaction share	−0.073**	−0.097***	−0.048
	(0.035)	(0.034)	(0.037)
Stable gap × gross-to-net reduction share	+0.214**	+0.156*	+0.276***
	(0.090)	(0.090)	(0.095)
Observations / date clusters			1,209 / 403
Asset and date effects			Yes
Origin-day activity controls			Yes

Note: Each cell reports the interaction of a 10 pp larger stablecoin route-minus-capital gap with a 10 pp larger V4 accounting proxy. Outcomes are future log admissible vehicle-token-side modify-liquidity flow over 120 days. The unit, fixed effects, controls, covariance, and stablecoin sample match Table 26. Add and remove flows value the vehicle-token sides assigned through V4 modify-liquidity records. These predictive specifications measure vehicle-token flows; whole-pool inventory and causal LP supply remain outside their scope. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 31: Stablecoin shortfalls and the V4-minus-V3 LP-action response

Outcome [log points]	7 days	30 days	120 days
LP actions	+0.163***	+0.212***	+0.320***
	(0.016)	(0.015)	(0.013)
Active origins	+0.045***	+0.062***	+0.115***
	(0.011)	(0.011)	(0.010)
Stacked observations / dates			5,198 / 522
Asset-date and protocol effects			Yes
Current protocol LP controls			Yes

Note: Each cell reports the V4-minus-V3 difference in the association between a 10 pp larger stablecoin route-minus-capital gap and the future LP-action outcome, with standard errors in parentheses. Outcomes are log points. The stacked unit is a protocol–vehicle–date row restricted to dates with observed V4 singleton swap activity. All columns absorb vehicle-date and protocol fixed effects, include same-protocol current LP-action and active-origin controls, and cluster by date. V3 outcomes are mint and burn event counts and active origins; V4 outcomes are modify-liquidity event counts and active origins. The estimates are conditional protocol differences and do not identify the effect of V4 adoption. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 32: Stablecoin shortfalls and the V4-minus-V3 LP-flow response

Outcome	7 days	30 days	120 days
Gross LP flow [log points]	−0.043	−0.009	+0.029
	(0.033)	(0.027)	(0.022)
Add-side flow [log points]	−0.043	−0.023	+0.012
	(0.033)	(0.027)	(0.023)
Remove-side flow [log points]	−0.032	+0.007	+0.046**
	(0.034)	(0.028)	(0.022)
Narrow/medium flow share [pp]	+1.530***	+1.126***	+0.329***
	(0.236)	(0.133)	(0.095)
Broad flow share [pp]	−1.530***	−1.126***	−0.329***
	(0.236)	(0.133)	(0.095)
Stacked observations / dates			5,188 / 521
Asset-date and protocol effects			Yes
Current protocol LP-flow controls			Yes

Note: Each cell reports the V4-minus-V3 difference in the association between a 10 pp larger stablecoin route-minus-capital gap and future vehicle-token-side LP flow, with standard errors in parentheses. Flow-amount outcomes are log points; range-composition outcomes are pp. The stacked unit is a protocol–vehicle–date row restricted to dates with observed V4 singleton swap activity. All columns absorb vehicle-date and protocol fixed effects, include same-protocol current gross, add, remove, and sender-day flow controls, and cluster by date. V3 outcomes value mint and burn token sides; V4 outcomes value modify-liquidity token sides. The estimates are conditional protocol differences and do not identify the effect of V4 adoption. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 33: Stablecoin shortfalls and the V4-minus-V3 reported-liquidity footprint

Horizon	Reported TVL growth [log points]	Pool-footprint growth [log points]
7 days	$+0.237^{***}$ (0.073)	$+0.075^{***}$ (0.017)
30 days	$+0.297^{***}$ (0.056)	$+0.149^{***}$ (0.020)
120 days	$+0.637^{***}$ (0.042)	$+0.275^{***}$ (0.022)
Obs. / date clusters (7d)		2,986 / 301
Obs. / date clusters (30d)		2,756 / 278
Obs. / date clusters (120d)		1,856 / 188
Asset-date and protocol effects		Yes
Origin TVL and pool-count controls		Yes

Note: Each cell reports the V4-minus-V3 difference in the association between a 10 pp larger stablecoin route-minus-capital gap and future growth in vehicle-linked reported full-pool TVL or pool count, with standard errors in parentheses. Outcomes are changes in log one plus the reported quantity. The stacked unit is a protocol–vehicle–date row restricted to V4-active dates with strict origin and target support in both protocols. All columns absorb vehicle-date and protocol fixed effects, include same-protocol origin TVL and pool-count controls, and cluster by date. A pool contributes its full reported TVL when it contains the vehicle; vehicle–vehicle pools can enter both vehicle series. The quantity is a reported stock-and-footprint proxy; reconstructed deposited capital, side-specific LP inventory, and causal protocol effects remain outside its scope. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

References

Adams, H., N. Zinsmeister, M. Salem, R. Keefer, and D. Robinson (2021). Uniswap v3 core. Technical whitepaper.

Amiti, M., O. Itskhoki, and J. Konings (2022). Dominant currencies: How firms choose currency invoicing and why it matters. *Quarterly Journal of Economics 137*(3), 1435–1493.

Bank for International Settlements Innovation Hub (2024). Nexus: Foreign exchange provision, key points. Project Nexus technical documentation. Accessed 20 August 2026.

Bank for International Settlements Innovation Hub (2025). Project rialto: Improving instant cross-border payments using central bank money settlement. Updated 10 December 2025.

Caparros, B., A. Chaudhary, and O. Klein (2024). Blockchain scaling and liquidity concentration on decentralized exchanges. Working paper, Warwick Business School and Gillmore Centre for Financial Technology.

Catalini, C., A. de Gortari, and N. Shah (2022). Some simple economics of stablecoins. *Annual Review of Financial Economics 14*, 117–135.

Chen, D. and D. Duffie (2021). Market fragmentation. *American Economic Review 111*(7), 2247–2274.

Daian, P., S. Goldfeder, T. Kell, Y. Li, X. Zhao, I. Bentov, L. Breidenbach, and A. Juels (2020). Flash boys 2.0: Frontrunning in decentralized exchanges, miner extractable value, and consensus instability. In *2020 IEEE Symposium on Security and Privacy*, pp. 910–927.

Dowd, K. and D. Greenaway (1993). Currency competition, network externalities and switching costs: Towards an alternative view of optimum currency areas. *The Economic Journal 103*(420), 1180–1189.

Eren, E. and S. Malamud (2022). Dominant currency debt. *Journal of Financial Economics 144*(2), 571–589.

Flandreau, M. and C. Jobst (2009). The empirics of international currencies: Network externalities, history and persistence. *The Economic Journal 119*(537), 643–664.

Gopinath, G., E. Boz, C. Casas, F. J. Díez, P.-O. Gourinchas, and M. Plagborg-Møller (2020). Dominant currency paradigm. *American Economic Review 110*(3), 677–719.

Gopinath, G. and J. C. Stein (2021). Banking, trade, and the making of a dominant currency. *Quarterly Journal of Economics 136*(2), 783–830.

Gorton, G. B. and J. Y. Zhang (2023). Taming wildcat stablecoins. *University of Chicago Law Review 90*(3), 909–971.

Kiyotaki, N. and R. Wright (1989, August). On money as a medium of exchange. *Journal of Political Economy 97*(4), 927–954.

Krugman, P. (1980). Vehicle currencies and the structure of international exchange. *Journal of Money, Credit and Banking 12*(3), 513–526.

Lehar, A. and C. A. Parlour (2025). Decentralized exchange: The uniswap automated market maker. *Journal of Finance 80*(1), 321–374.

Lyons, R. K. and G. Viswanath-Natraj (2023). What keeps stablecoins stable? *Journal of International Money and Finance 131*, 102777.

Makarov, I. and A. Schoar (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics 135*(2), 293–319.

Milionis, J., C. C. Moallemi, T. Roughgarden, and A. L. Zhang (2022). Automated market making and loss-versus-rebalancing.

Mukhin, D. (2022). An equilibrium model of the international price system. *American Economic Review 112*(2), 650–688.

Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. *Federal Reserve Bank of St. Louis Review 103*(2), 153–174.

Somogyi, F. (2026). Dollar dominance in FX trading. *Management Science*.

Xi, W. and C. C. Moallemi (2026). Quantifying sub-optimality in routing for automated market makers. Accepted at the 5th Workshop on Decentralized Finance.